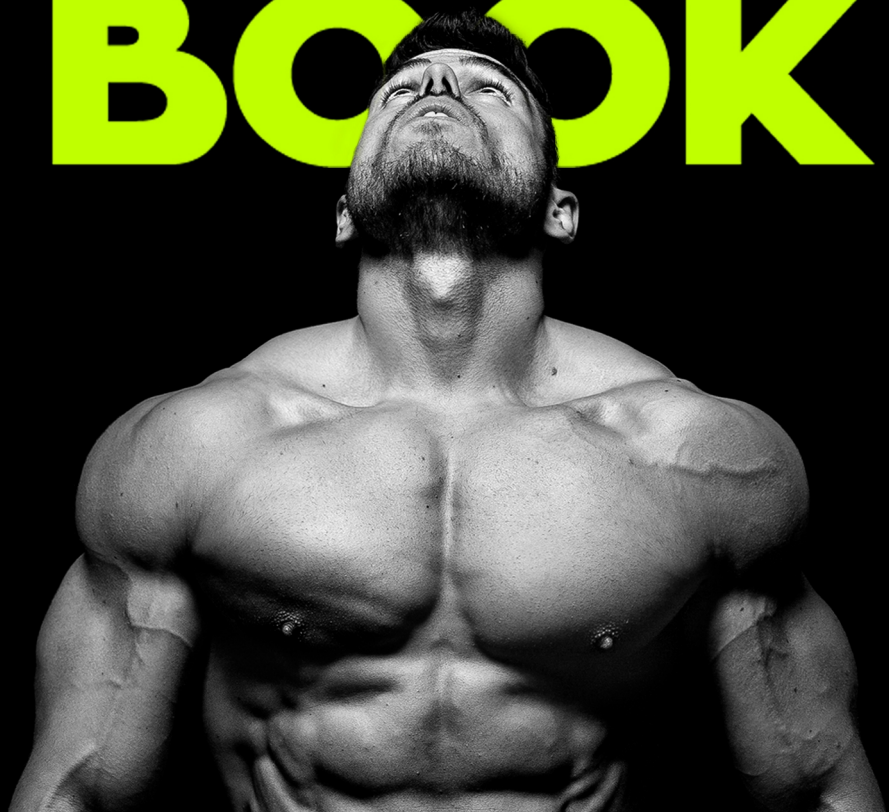
How to Train for Anything & Everything, Anywhere & Everywhere

ROSS EDGLEY

THE WORLD'S FITTEST BOOK



LIVE BEYOND REPS AND CALORIES

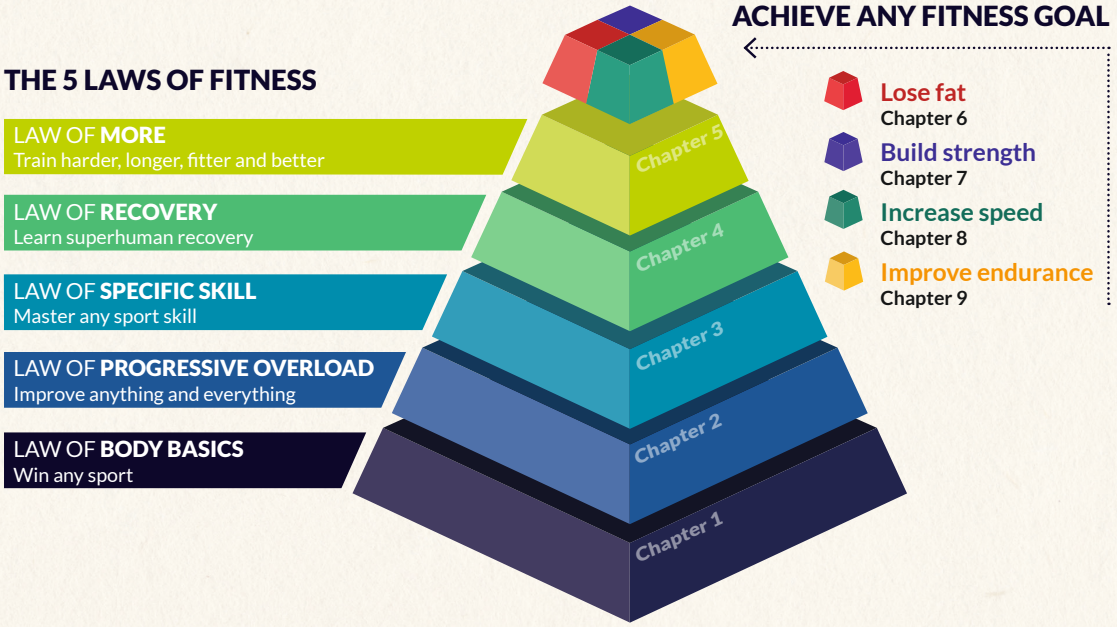
Note the word **average**.

Variable	Strength	Power	Hypertrophy	Endurance
Load (% of one-rep maximum*)	80–100	70–100	60–80	40–60
Repetitions per set	2–5	1–5	6–15	25–60
Sets per exercise	4–7	3–5	4–8	2–4
Rest between sets (minutes)	2–6	2–6	2–5	1–2
Durations (seconds per set)	5–10	4–8	20–60	80–150
Speed per rep (% of max)	60–100	90–100	60–90	60–80
Training sessions per week	3–6	3–6	5–7	8–14

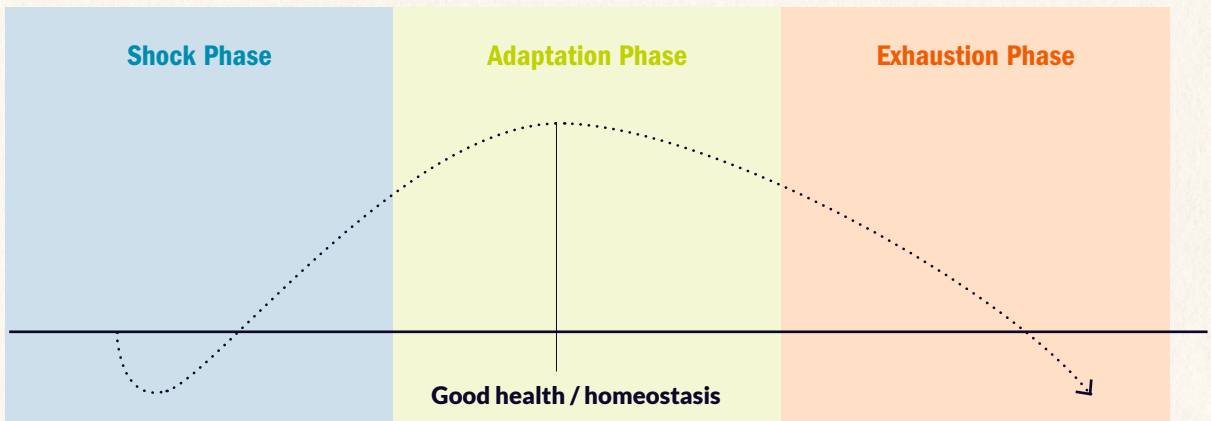
* 'One rep maximum' (1RM) is a term used to define the maximum amount of force that can be generated in one maximal contraction or lift.

HEALING:
PART I, MY BEGINNING

Pyramids of Priority



SELYE'S GENERAL ADAPTION SYNDROME



The perfect adaptation is when you train to maximise the time spent in the Adaptation Phase, while avoiding any time spent in the Exhaustion Phase.

PART II:
THE FIVE LAWS OF FITNESS

THE WORLD'S FIRST TREE-ATHLON | THE 12-WEEK PLAN

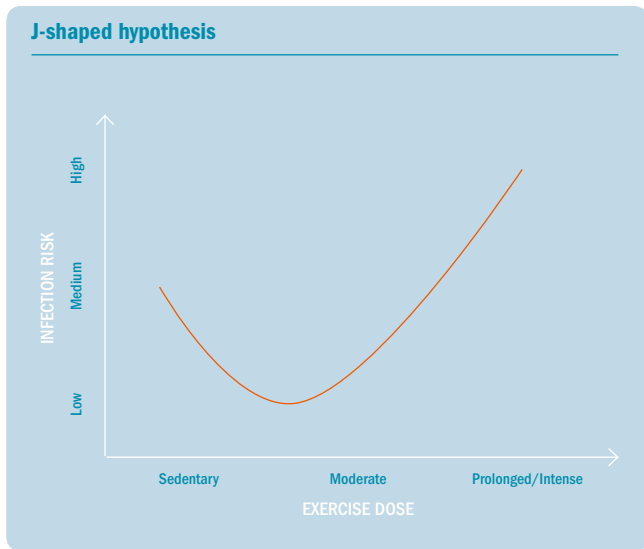
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 1	**Swimming specific** · 10 minutes warm-up - 10 x 25m drills with 20 seconds rest · 10 x 50m (25m hard/25m easy) with 20 seconds rest · 5 minutes cool -down	**Running specific** · 10 minutes warm-up · 6 x 20 seconds hill reps and jog down recovery · 15 minutes running at 80% max. heart rate · 5 minutes cool-down **Strength session** 30 minutes	**Rest day**	**Cycling specific** · Turbo 40 minutes (10 minutes warm-up. 10 x 60 seconds high cadence (low gear at <75% max. heart rate) and 60 seconds easy. 10 minutes cool-down.)	**Swimming specific** · 30–45 minutes technique work	**Running specific** · 40 minutes easy and off-road · Strength session 30 minutes	**Cycling specific** · 60 minutes steady pace
Week 2	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with 20 seconds rest · 15 x 50m (25m hard/25m easy) with 20 seconds rest · 5 minutes cool-down	**Running specific** · 10 minutes warm-up · 8 x 20 seconds hill reps and jog down recovery · 15 minutes running at 80% max. heart rate · 5 minutes cool-down **Strength session** 30 minutes	**Rest day**	**Cycling specific** · Turbo 45 minutes (10 minutes warm-up. 12 x 60 seconds high cadence (low gear at <75% max. heart rate) and 60 seconds easy. 10 minutes cool-down.)	**Swimming specific** · 30–45 minutes technique work	**Running specific** · 45 minutes easy and off-road **Strength session** · 30 minutes	**Cycling specific** · 70 minutes steady pace
Week 3	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with 20 seconds rest · 20 x 50m (25m hard/25m easy) with 20 seconds rest · 5 minutes cool-down	**Running Specific** · 10 minutes warm-up · 10 x 20 seconds hill reps with jog down recovery · 15 minutes running at 80% max. heartrate · 5 minutes cool-down **Strength session** 30 minutes	**Rest day**	**Cycling specific** · Turbo 40 minutes (10 seconds warm-up, 15 x 60 seconds high cadence (low gear at <75% max. heart rate) and 60 seconds easy. 10 minutes cool-down.)	**Swimming specific** · 30–45 minutes technique work	**Running specific** · 50 minutes easy and off-road **Strength session** · 30 minutes	**Cycling specific** · 80 minutes steady pace

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 4	**Swimming specific** · Complete 50% of race distance non-stop	**Running specific** · 10 minutes warm-up · 15 x 20 seconds brisk strides at 5k pace and 40 seconds recovery · 5 minutes cool-down **Weights** · 30 minutes	**Rest day**	**Cycling specific** · Turbo 25 minutes (10 minutes warm-up. 10 x 30 seconds at high cadence (low gear <75% max. heart rate) and 30 seconds recovery. 5 minutes cool-down.)	**Swimming specific** · 30 minutes technique work	**Running specific** · 30 minutes easy and off-road **Strength session** · 30 minutes	**Cycling specific** · 45 minutes steady pace
Week 5	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with 20 seconds rest · 10 x 100m (50m hard/ 50m easy) with 20 seconds rest · 5 minute cool-down	**Running specific** · 10 minutes warm-up · 6 x 30 seconds hill reps with jog down recovery · 15 seconds running at 80-85% max. heart rate · 5 minutes cool-down **Strength session** · 30 minutes	**Rest day**	**Cycling specific** · Turbo 40 minutes (10 minutes warm-up. 10 x 30 seconds at max intensity. 90 seconds easy. 10 minutes cool-down.)	**Swimming specific** · 45–60 minutes technique and endurance work	**Running specific** · 45 minutes easy and off-road. Find a hilly route to build leg strength	**Cycling specific** · 70 minutes hilly at steady pace. Stay seated on hills to build leg strength
Week 6	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with 20 seconds rest · 15 x 50m (25m hard/25m easy) with 20 seconds rest · 5 minutes cool-down	**Running Specific** · 10 minutes warm-up · 8 x 20 seconds hill reps and jog down recovery · 15 minutes running at 80% max. heart rate · 5 minutes cool-down **Strength session** · 30 minutes	**Rest day**	**Cycling specific** · Turbo 45 minutes (10 minutes warm-up. 12 x 60 seconds high cadence (low gear at <75% max. heart rate) and 60 seconds easy. 10 minutes cool-down.)	**Swimming specific** · 30–45 minutes technique work	**Running specific** · 45 minutes easy and off-road **Strength session** · 30 minutes	**Cycling specific** · 70 minutes steady pace

	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 7	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with - 20 seconds rest · 20 x 50m (25m hard/25m easy) with 20 seconds rest · 5 minutes cool-down	**Running specific** · 10 minutes warm-up · 10 x 20 seconds hill reps with jog down recovery · 15 minutes running at 80% max. heart rate · 5 minutes cool-down **Strength session** · 30 minutes	**Rest day**	**Cycling specific** · Turbo 40 minutes. (10 seconds warm-up. 15 x 60 seconds high cadence (low gear at <75% max. heart rate) and 60 seconds easy. 10 minutes cool-down.)	**Swimming specific** · 30–45 minutes technique work	**Running specific** · 50 minutes easy and off-road **Strength session** · 30 minutes	**Cycling specific** · 80 minutes steady pace
Week 8	**Swimming specific** · Complete 50% of race distance non-stop	**Running specific** · 10 minutes warm-up · 15 x 20 seconds brisk strides at 5k pace and 40 seconds recovery · 5 minutes cool-down **Strength session** · 30 minutes	**Rest day**	**Cycling Specific** · 25 minutes turbo. (10 minutes warm-up. 10 x 30 seconds at high cadence (low gear <75% max. heart rate) and 30 seconds recovery. 5 minutes cool-down.)	**Swimming specific** · 30 minutes technique work	**Running specific** · 30 minutes easy and off-road · Strength session 30 minutes	**Cycling specific** · 45 minutes steady pace
Week 9	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with - 20 seconds rest · 6 x 100m hard with 30 seconds rest · 5 minutes cool-down	**Running specific** · 10 minutes warm-up · 6 x 30 seconds hill reps and jog down recovery · 2 x 5 minutes running at 85–90% max. heart rate + 2 minutes active recovery · 5 minutes cool-down **Strength session** · 30 minutes	**Rest day**	**Cycling specific** · Turbo 45 minutes (10 minutes warm-up. 8 x 2 minutes at 85–90% max. heart rate and 60 seconds easy spin with 10 minutes cool-down.) · 5 minutes run	**Swimming specific** · 45–60 minutes, open water if possible. If not, include open-water skills	**Running specific** · 55 minutes off-road and easy. Run on terrain similar to race route **Strength session** · 30 minutes	**Cycling specific** · 70 minutes including 10 minutes at goal race pace

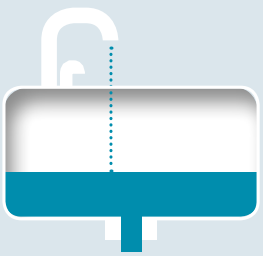
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Week 10	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with 20 seconds rest · 8 x 100m hard with 30 seconds rest after each 100m · 5 minutes cool-down	**Running specific** · 10 minutes warm-up · 8 x 30 seconds hill reps and jog down recovery · 2 x 8 minutes running at 85–90% max. heart rate with 2 minutes active recovery · 5 minutes cool-down **Strength session** · 30 minutes	**Rest day**	**Cycling specific** · Turbo 45 minutes (10 minutes warm-up. 5 x 3 minutes at 85–90% max. heart rate with 90 seconds easy spin and 10 minutes cool-down.) · 10 minutes run	**Swimming specific** · 45–60 minutes open water if possible. If not, include open-water skills	**Running specific** · 55 minutes off-road and easy. Run on terrain similar to race route **Strength session** · 30 minutes	**Cycling specific** · 80 minutes including 15 minutes at goal race pace
Week 11	**Swimming specific** · 10 minutes warm-up · 10 x 25m drills with 20 seconds rest · 10 x 100m hard + 30 seconds rest · 5 minutes cool-down	**Running specific** · 10 minutes warm-up · 10 x 30 seconds hill reps and jog down recovery · 15 minutes running at - 85–90% max. heart rate · 5 minutes cool-down **Strength session** · 30 minutes	**Rest day**	**Cycling specific** · Turbo 45 minutes (10 minutes warm-up. 3 x 5 minutes at 85–90% max. heart rate and 2 minutes easy spin. 10 minutes cool-down.) · 15 minutes run	**Swimming specific** · 45–60 minutes open water if possible. If not, include open-water skills	**Running specific** · 40 minutes off road. Run on terrain similar to race route **Strength session** · 30 minutes	**Cycling specific** · 90 minutes including middle 20 minutes at goal race pace
Week 12	**Swimming specific** · 10 minutes warm-up · 10 x 50m drills with 20 seconds rest · 6 x 50m sprints at goal race pace and 30 seconds rest · 5 minutes easy cool-down	**Rest day**	**Cycling specific** · 15 minutes warm-up · 15 minutes sustained effort at goal race pace **Running specific** · 5 minutes at goal race pace and 5 minutes easy jog to cool down	**Rest day**	**Swimming specific** · 15 minutes including 5 x 25-50m sprints	**Cycling specific** · 15 minutes including middle 5 minutes at goal race pace **Running specific** · 5 minutes easy pace off the bike	**Race day**

EXCESSIVE EXERCISE AND IMMUNITY: THE J-SHAPED CURVE

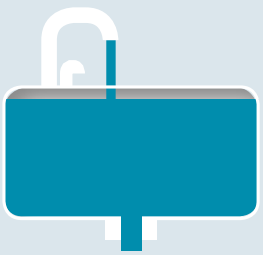


The Active Female by Chamorro-Viña, Carolina, Maria Fernandez-del-Valle and Anna M. Tacón (2013)

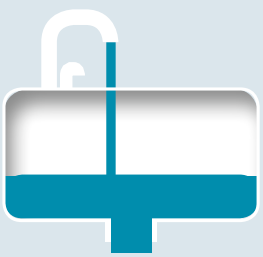
Work capacity and recovery: The 'sink' analogy



LOW STRESS, BUT SUFFICIENT RECOVERY



HIGH STRESS AND INSUFFICIENT
RECOVERY DUE TO LOW WORK CAPACITY



HIGH STRESS BUT SUFFICIENT RECOVERY
BECAUSE OF IMPROVED WORK CAPACITY

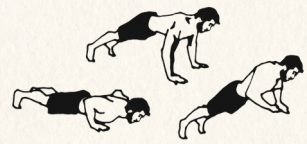
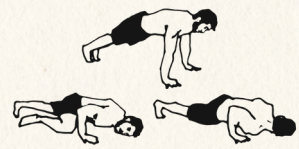
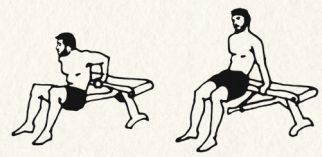
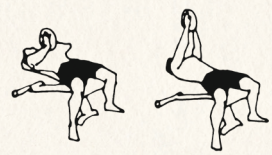
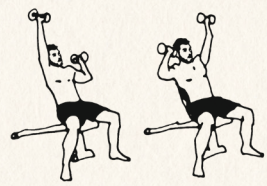
The World's Easiest Kettlebell Workout!

Add a rep each session...
Session 1: 5 reps x 5 sets
Session 2: 6 reps x 5 sets
Session 3: 7 reps x 5 sets
Session 4:: 8 reps x 5 sets
Session 5: 9 reps x 5 sets
Session 6: 10 reps x 5 sets (now you're doing 50 swings in 5 minutes)
Once you get to 10 reps for five sets start adding a set each session
Session 7: 10 reps x 6 sets
Session 8: 10 reps x 7 sets
Session 9: 10 reps x 8 sets
Session 10: 10 reps x 9 sets
Session 11: 10 reps x 10 sets (now you're doing 100 swings in 10 minutes)

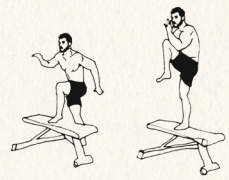
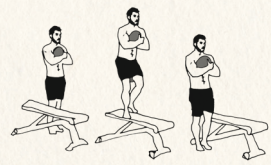
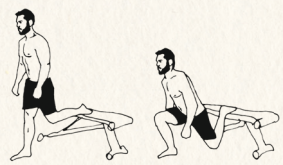
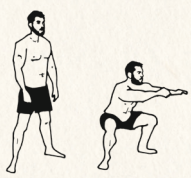
YOUR 12-WEEK WORLDWIDE BODYWEIGHT WORKOUT

	4-DAY ROUTINE	5-DAY ROUTINE	6-DAY ROUTINE
	WEEKS 1–4	**WEEKS 5–8**	**WEEKS 9–12**
KEY FOCUS	Improve strength Refine technique Develop kinaesthetic awareness and better mechanical energy	Improve strength Refine technique Increase work capacity	Improve strength Refine technique Specialise
WORKOUT	**Monday:** 'Push' Training **Tuesday:** Lower-body Training **Wednesday:** REST **Thursday:** 'Pull' Training **Friday:** Full-body Centurion Workout **Saturday:** REST **Sunday:** REST	**Monday:** 'Push' Training **Tuesday:** Lower-body Training **Wednesday:** 'Pull' Training **Thursday:** REST **Friday:** Full-body Caveman Conditioning **Saturday:** Ab Workout **Sunday:** REST	**Monday:** 'Push' Training **Tuesday:** Lower-body Training **Wednesday:** 'Pull' Training **Thursday:** Ab Workout **Friday:** Full-body Caveman Conditioning **Saturday:** Sport Specific **Sunday:** REST
MAIN GOAL	Unlock the neurological potential of the body and develop kinaesthetic awareness.	Increase work capacity and therefore your body's ability to tolerate more training.	Use your improved neurological potential and work capacity to specialise 1 day per week.

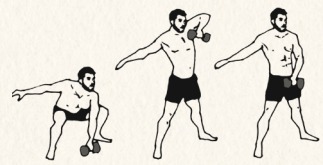
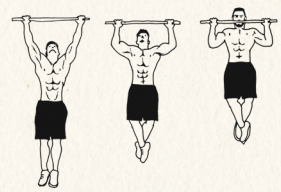
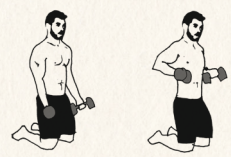
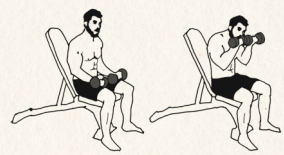
4-DAY ROUTINE | WEEKS 1–4 | MONDAY: 'PUSH' TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Plyometric Push-up	5	3	90 secs	Get in a press-up position. Hands shoulder-width apart and back straight. Lower your chest to the floor, then push up explosively until your arms fully extend. Attempt to add claps into the movement while airborne if possible. (If you can't manage full press-ups, do them on your knees.)	
Spiderman Push-up	5 (each leg)	3	90 secs	Get into a traditional press-up position. Lower yourself toward the floor and bring your right knee to your right elbow, keeping it off the ground. Press back up and return your leg to the starting position. Repeat with the alternate leg. (If you can't manage full press-ups, do them with your hands on a bench.)	
Bench Dip	12	3	60 secs	Stand facing away from a bench. Grab it with both hands at shoulder width. Extend your legs out in front of you. Slowly lower your body by flexing at the elbows until your upper arms create a 90-degree angle. Use your triceps to lift yourself back to the starting position.	
Plate Press	20	3	60 secs	Lie on a flat bench holding a weight plate over your chest with your palms facing each other. Pinching and squeezing the plate, push up until your arms are straight, pause, then lower under control.	
Alternate Dumbbell Press	12 (each arm)	3	60 secs	Hold a dumbbell in each hand just above shoulder height with your palms facing forward and arms bent. Press one dumbbell up over your head without moving the other dumbbell. Return the weight to the starting position. Repeat for the opposite side, continuing to alternate between arms.	
Bear Crawl	10m forward then 10m backward	3	45 secs	Get down on all fours, with your hands directly under your shoulders and your knees under your hips. Bring your knees off the ground and travel forward. Keep your back flat at all times. Travel 10 m forward, then reverse.	

4-DAY ROUTINE | WEEKS 1–4 | TUESDAY: LOWER-BODY TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Explosive Sprinter Step-up	5 (each leg)	3	90 secs	Place a bench or box in front of you and step onto it with one foot. As you plant your foot, aggressively drive through your other foot, bringing your knee as high as you can. Lower it back down to step back onto the floor. Repeat on the other side.	
Lateral Step-over with Weighted Plate	10 (each leg)	3	90 secs	Holding a weight plate at your chest, stand to the right side of the bench and place your left foot on top. Push from your left heel and raise your body onto the bench. Touch down on the bench with your right foot, then lower your left foot to the other side of the bench. Drive back up and repeat on the other side.	
Bulgarian Split Squat	12 (each leg)	3	60 secs	Stand facing away from the bench. Rest one leg on the bench behind you with the laces of your shoes facing down. Squat with your standing leg until the knee of your trailing leg almost touches the floor. Push up through the front foot to return to the start position.	
Paused Bodyweight Squat	20	5	60 secs	Stand with your feet shoulder-width apart. Bend the knees and sit back with your hips until your knees are at a 90-degree angle. Pause, then reverse the motion until you return to the starting position.	
Duck Walk	5m forward then 5m backward	4	60 secs	Lower into a squat so your thighs are parallel to the floor and place your hands behind your head. Slowly walk forward 5 m in a controlled manner, keeping your balance at all times, then reverse the movement back to the start position.	

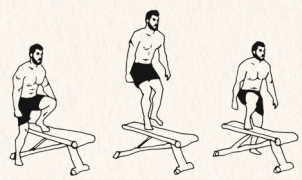
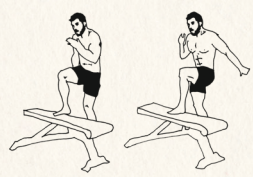
4-DAY ROUTINE | WEEKS 1–4 | THURSDAY: 'PULL' TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Dumbbell Single Arm Sumo High Pull	5 (each arm)	3	90 secs	Stand with your legs wider than shoulder-width apart with a dumbbell in one hand. Lower your body into a squat position until your thighs are parallel to the floor. Push through your heels and raise the dumbbell to your chin, so your arm bends and the elbow is in line with the top of your head. Squat down and repeat.	
Pull-up	8	3	90 secs	Grip the bar with hands shoulder-width apart. Raise your feet off the floor and hang. Pull yourself up until your chin is over the bar and repeat.	
Incline Dumbbell Row	12	3	60 secs	Holding a dumbbell in each hand, lie face-down on an incline bench. Keep your core tight and back straight as you row the weights up to your chest. Lower and repeat.	
Reverse Incline Scaption	20	3	60 secs	Lie face-down on an incline bench holding 2 dumbbells by your sides. Arc the weights up to touch above your head, keeping your arms straight at all times. You should feel a strong stretch across your shoulders. Return slowly to the start position.	
Kneeling Drag Bicep Curl	10	4	60 secs	On your knees with your back straight, hold a dumbbell in each hand in front of your thighs with an underhand, shoulder-width grip. Curl the weights and, as your hands rise, drive the elbows back so the dumbbells stay as close to your torso as possible. Squeeze your biceps at the top and then lower the weight under control.	
Paused Seated Dumbbell Curl	12	4	60 secs	In a seated position, grab a pair of dumbbells with palms facing out. Raise the dumbbells on both sides so that the elbow is flexed at a 90-degree angle. Curl both dumbbells up to the nose. Pause for 1 second, then return to the 90-degree angle. The beauty of this exercise is the biceps are constantly in tension.	

4-DAY ROUTINE | WEEKS 1–4 | FRIDAY: FULL-BODY CENTURION WORKOUT

This fitness test is simple enough. Created by myself and *Men's Health* to mass-test the fitness of millions who downloaded the Primal 9 training system, it gets you to do 25 reps of four moves – 100 reps – without a break. Then you rest. Then you do another round. And another. And another. You're aiming for 10 rounds – and 1000 reps in total – to complete the whole thing. I know. Sounds impossible and very few will be able to finish it without some large tactical rest breaks. But that's the point.

Get as far through The Centurion as you can now and – trust me – once you progress through the 12-week programme, you'll soon be storming through it since the improvements are both biomechanical (better movement) and physiological (enhanced endurance).

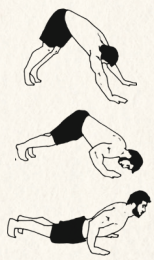
Exercise	Reps	Coaching Cues	
Bench Jump-over	25	Stand next to the bench and grasp it with both hands. Jump over with both feet while keeping hold of the bench, landing on the other side. Jump back over and continue hopping from side to side, landing lightly.	
Lateral Step-over	25	Stand to the right side of the bench and place your left foot on top. Push from your left heel and raise your body onto the bench. Touch down on the bench with your right foot, while at the same time lowering your left foot to the other side of the bench. Drive back up and repeat on the other side.	
Bear Crunch	25	Assume a press-up position. Crunching your core as you turn, bring your left knee towards your right hip. Repeat on the other side.	
Bench Toe Tap	25	Stand with one foot on a bench or high step. Jump and swap your feet over, tapping the bench lightly with your toe.	

That's one round.

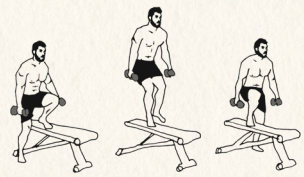
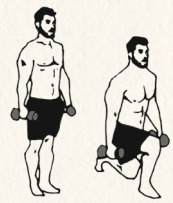
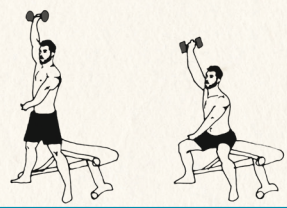
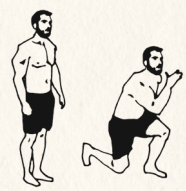
Rest as long as you need, then do another.

Complete 10 rounds. Note down your time. That's your Centurion personal best.

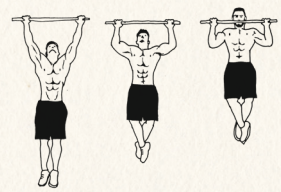
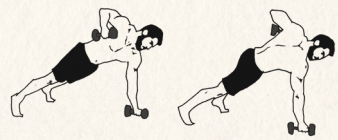
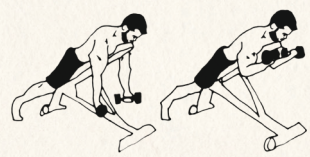
5-DAY ROUTINE | WEEKS 5–8 | MONDAY: 'PUSH' TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Plyometric Push-up	5	3	90 secs	Get in a press-up position, hands shoulder-width apart and back straight. Lower your chest to the floor, then push up explosively until your arms fully extend. Attempt to add claps into the movement while airborne if possible. (If you can't manage full press-ups, do them on your knees.)	
Hindu Push-up	10	3	90 secs	Get in a push-up position. Keep your core tight and back flat. Push your torso backward and raise your bum up to move into a pike position so your body forms an inverted V. Bend your elbows to lower your upper chest to the ground while keeping your bum in the air. As you lower your chest, drop your bum down so your body is in a straight line when you're closest to the ground. Straighten your arms to push your chest up, but allow for a slight arch in your back similar to the yoga Upward Dog position. From this position, push your torso and repeat.	
Dumbbell Fly into Hex Press	12	3	60 secs	Lie flat on the bench holding 2 dumbbells above your chest with your palms facing each other. With a slight bend at your elbows, lower the dumbbells down to your sides until you feel your chest stretch. Bring the dumbbells into the centre of your chest and push them back up to the start position and repeat.	
Arnold Press	12	3	60 secs	Grab a set of dumbbells and bring them to shoulder height with arms bent and palms facing the body. Gripping the dumbbells as tightly as possible, press them overhead, rotating the dumbbells until the palms are facing forward at the top of the press. Pause, and then slowly lower the weight, reversing the rotating so that the palms end facing the body at the bottom of the press.	
Rolling Overhead Tricep Extension	12	3	60 secs	Lie on a bench with both dumbbells extended overhead. Slowly bend at the elbow to lower the dumbbells behind your head, resisting gravity as much as possible. When your forearms are parallel to the bench, bring the dumbbells back to your chest and push them up. Repeat.	
Lizard Walk	5m forward then 5m backward	3	45 secs	Get into a plank position. Take your first step by bringing your left hand and right knee forward, coming down into a press-up position. Repeat on the other side. Walk 5 m forward, then reverse the movement 5 m backward to the start position.	

5-DAY ROUTINE | WEEKS 5–8 | TUESDAY: LOWER BODY TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Explosive Lunge	5 (each leg)	3	90 secs	Lunge forward until your back knee is almost touching the ground. Lunge into the air, bringing your rear foot forward and the front foot back. Land in a lunge position and repeat.	
Dumbbell Lateral Step-over	10 (each leg)	3	90 secs	Holding a dumbbell in either hand, stand to the right side of the bench and place your left foot on top. Push from your left heel and raise your body onto the bench. Touch down on the bench with your right foot, then lower your left foot to the other side of the bench. Drive back up and repeat on the other side.	
Weighted Lunge	10m (forward)	3	60 secs	Stand with your torso upright, holding 1 dumbbell in either hand by your sides. Take a big step forward with your right leg and start to shift weight forward so your heel hits the floor first. Lower your body until your right thigh is parallel to the floor and your right shin is vertical (it's OK if your knee shifts forward a little as long as it doesn't go past your right toe). If mobility allows, lightly tap your left knee to the ground while keeping the weight in the right heel. Press into your right heel to drive back up to starting position. Repeat on the other side.	
Single Overhead Dumbbell Box Squat	10 (each leg)	5	60 secs	Hold a dumbbell in one hand. Raise it straight overhead, lock the elbow and keep it there for the duration of the set. Use your nonworking hand to balance. Keeping the abs tight, begin squatting so your bum touches the bench and then return to the starting position.	
Reverse Lunge	10m (backward)	4	60 secs	Take a large and controlled step backward with your left foot. Lower your hips so that your right thigh (front leg) becomes parallel to the floor with your right knee positioned directly over your ankle. Your left knee should be bent at a 90-degree angle and pointing toward the floor with your left heel lifted. Return to standing by pressing your right heel into the floor and bringing your left leg forward to complete one rep. Alternate legs and step back with your right leg.	

5-DAY ROUTINE | WEEKS 5–8 | WEDNESDAY: PULL TRAINING

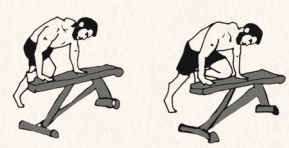
Exercise	Reps	Sets	Rest	Coaching Cues	
Paused Explosive Dumbbell Deadlift	10	3	90 secs	The paused deadlift is brutal. Essentially, it's a standard deadlift from the floor, but instead of pulling continuously until lockout, you pause below the knees around mid-shin level for a 2–3 second count. This increases strength and time under tension, which improves overall muscle growth.	
Pull-up	8	3	90 secs	Grip the bar with hands shoulder-width apart. Raise your feet off the floor and hang. Pull yourself up until your chin is over the bar and repeat.	
Renegade Row	10 (each arm)	3	60 secs	Get into a press-up position with your hands on the handles of 2 dumbbells. Keeping your core tensed, row the right dumbbell up to your abs and then return to the start position. Repeat with the left dumbbell to complete one repetition.	
Spider Dumbbell Curl	10	4	60 secs	Lie on an incline bench and hold a dumbbell in each hand. Let them hang underneath your shoulders. Use your biceps to curl the dumbbells towards your shoulders. Slowly return to the starting position.	
One-arm Plank Dumbbell Bicep Curl	12 (each arm)	4	60 secs	Place a dumbbell on the ground and set up in the up position of a press-up. Keep your feet shoulder-width apart. Grip the dumbbell in your right hand. Complete a bicep curl with your right hand by flexing at the elbow and curling upwards towards your shoulder. Lower your arm by fully extending it to the starting position. After 6 repetitions, repeat with the other arm. That's one set.	

5-DAY ROUTINE | WEEKS 5–8 | FRIDAY: FULL-BODY CAVEMAN CONDITIONING

Presenting The Mountain. The most popular full-body caveman conditioning circuit that was downloaded 1.2 million times, you might finish this quickly, or it might take a very long time indeed. Just make sure you complete the 10 rounds. Easier said than done. For your first session of Caveman Conditioning you're going to spend 20 seconds doing each Mountain Climber at varying angles. You can only rest after finishing your hands-elevated Mountain Climbers.

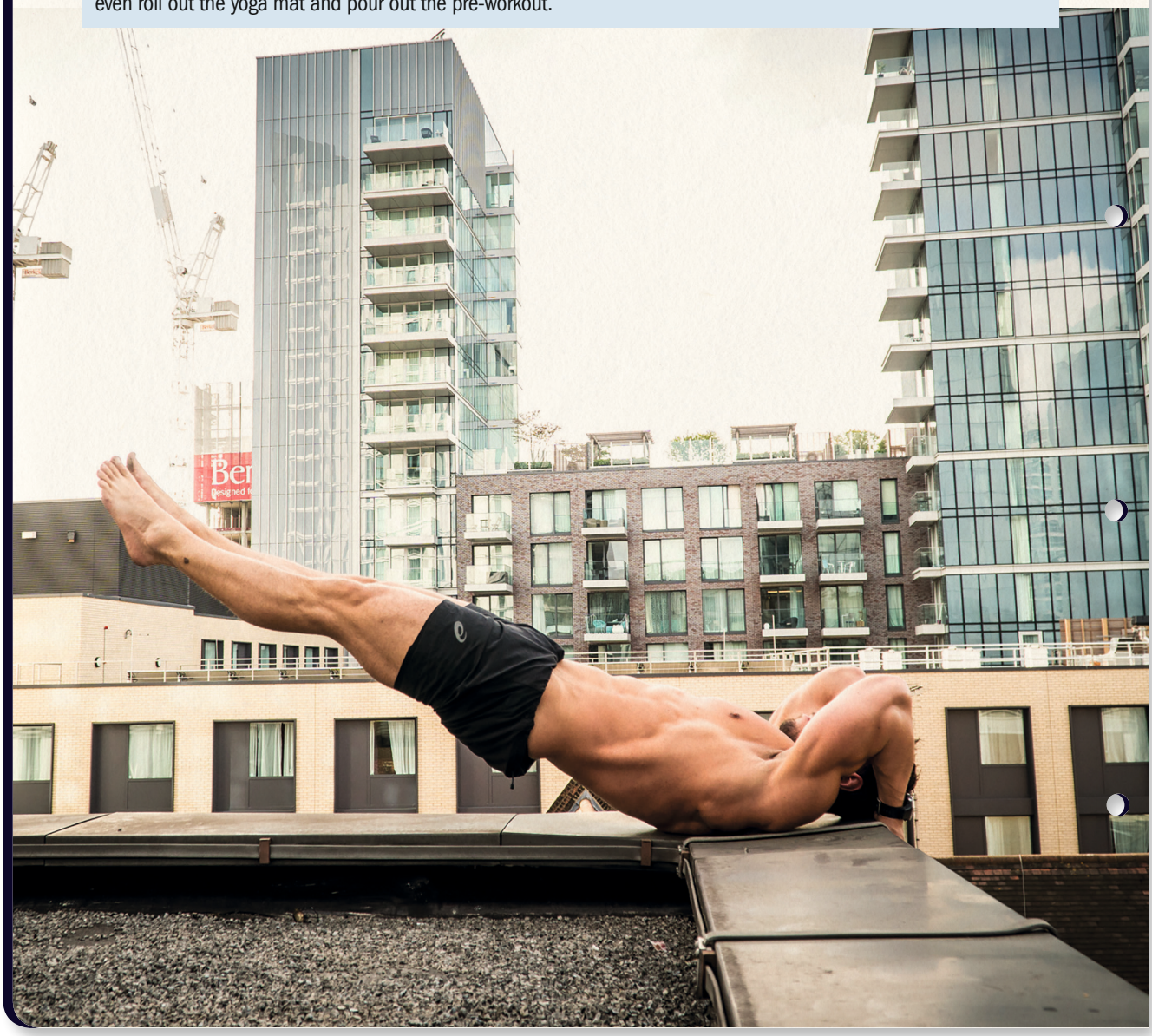
Repeat this process 10 times.

If you need more rest, you can take it. But you've got to make it through those 10 rounds.

Exercise	Reps	Sets	Rest	Coaching Cues	
Feet Elevated Mountain Climber	20 secs	10	n/a	Set up on the floor as though in a sprinter's blocks, with your feet resting on a bench. Explosively piston your knees in towards your chest. Repeat for 20 seconds. Go straight into the next move without rest.	
Mountain Climber	20 secs	10	n/a	Set up on the floor as though in a sprinter's blocks. Explosively piston your knees in towards your chest. Repeat for 20 seconds. Go straight into the next move without rest.	
Hands Elevated Mountain Climber	20 secs	10	60 secs	Set up on the floor as though in a sprinter's blocks with your hands resting on a bench. Explosively piston your knees in towards your chest. Repeat for 20 seconds. Then rest for 30 seconds (enjoy it) before starting another round of the three moves. Do 10 rounds.	

This ab workout is very different to most.

Placing a greater emphasis on large, full-body movements that produce more muscle activation in the abdominal area, it's been designed to condition and create a strong, functional and aesthetic core. All tried, tested and based on the findings from years of research within the world of 'six-pack sports science', which is why instead of listing exercises for you to blindly follow, I will explain the theory behind this workout (and all others in this book) before we even roll out the yoga mat and pour out the pre-workout.



There's NO perfect six-pack guide...

There's no single best stomach exercise.

Why? Because research in *The Journal of Manipulative and Physiological Therapeutics* conducted a large-scale literary review of 87 core-related studies only to find 'Overall, the studies retrieved lacked consistency, which made it impossible to extract aggregate estimates' and find the best ab routine.

Enter the researchers from Department of Kinesiology at The Pennsylvania State University, USA. They wanted to specifically challenge the crunches dominance as the best 'six-pack' exercise and decided to monitor which techniques activated the muscles the best. Large, integration movements like the plank that require different parts of the body to work in harmony. Or smaller, isolation movements like the standard crunch that only target the muscles of the stomach.

The results were as follows...

Increase your 'ab activation' by 27%

Sports scientists took a group of athletes.

Each was made to complete a series of core exercises while they were connected to surface electromyography (EMG) electrodes to monitor muscle activation. What did they discover? Activation of the abdominal muscles was greatest during larger, compound, integration movements. Not even a little, but a lot greater!

The rectus abdominus (the muscles of the 'six-pack') and external obliques (the muscles found down the side of the stomach) showed 27% greater activation during the 'plank with reach' compared to a traditional crunch. Activity in the lumbar erector spinae (the muscles of the back and spine) was also two times greater during the 'plank with reach' compared to the crunch. The 'tripod stance' of Mountain Climbers 'results in a greater activity for all the muscles'. The examples go on.

Get strong, functional 'compound' abs

These basic principles have been echoed and practised in sports for years. (This is especially true in gymnastics.)

'This extreme difference is an illustration of how the integration exercises may be a superior choice for a training regimen as they target a wider range of muscles for a more comprehensive strengthening effect.

JOURNAL OF STRENGTH AND CONDITIONING RESEARCH

In summary, results from the Pennsylvania State University study clearly show there was greater activation in the stomach muscles during big and integrative movements. That's not to say isolation movements aren't without their benefits, but it does mean that a core routine that is heavily based around crunches and sit-up variations might need to be redesigned.

Enter The Four Key Principles of Ab Training that each get progressively harder...

Core Principle 1: Learn to balance

Simply balancing can build your core.

This is based on research from *The Journal of Human Kinetics* that showed something as simple as performing your push-ups (or planks) with your feet placed in gymnastic rings elicited a different kind of muscle activation in the stomach when compared to a traditional push-up on the floor. 'Therefore, suspension push-ups may be considered an advanced variation of a traditional push-up when a greater challenge is warranted.'

They're not the only ones to think so either.

The Journal of Exercise Physiology states 'A suspension device elicited a greater activation of the stomach muscles.' Again supporting this idea to take your planks and press-up variations off the floor and onto something less stable like an exercise Swiss ball.

It's worth noting that unstable surface training still divides the strength and conditioning community and research published by the NSCA stated its application is 'still very limited'. But it is acknowledged that 'unstable surface training has been shown to increase core muscle activity and alter neuromuscular recruitment patterns'.

So, if your goal is to build a stronger core: learn to balance, alter your neuromuscular recruitment patterns, shock the abs into working over a different range of motion and add some variety to your push-ups.

How to perform 'T' push-ups

- Assume a standard push-up position, your body aligned from ankles to head.
- Lower your body until your chest nearly touches the floor.
- As you push yourself back up, rotate one side of your body up as you raise the same side arm your towards the ceiling.
- Pause, and return to the starting position and perform another push-up.
- Alternate sides with each rep.

Core Principle 2: Learn to hold

Hold and do nothing. This is one of the best ways to strengthen your stomach. This is because most strength and conditioning coaches will agree that L-sit variations and all kinds of holds will tax your stomach far more than any crunch or sit-up. If you've ever attempted hanging from a bar with a weight between your legs you'll know exactly what I mean.

This is, again, supported by research published by the American Association for Health, Physical Education and Recreation that found 'The exercises found most effective in strengthening abdominal musculature as revealed by the high magnitude of action potential are: L-sits and V-sits.'

How to L-sit

- Hang from a bar or rings.
- Place your hands around shoulder-width apart.
- Keep your legs straight and together.
- Using your lower abs and hip flexors, lift your legs.
- Once parallel to the floor, hold.
- Hold your legs at a 90° angle to your torso. This is the L-sit position.
- Once you've mastered this, try adding weight.

What if you can't L-sit? No problem.

Researchers from the University of Jyväskylä, in Finland, showed you don't have to in order to activate your core. They tested the effect isometric upper-body exercises had on the muscles of the stomach. Put more simply, they wanted to see if upper-body exercises where you hold a static position could engage the abdominal muscles.

What they found was that these exercises, like the Pallof Hold, elicited a, 'sufficient level of contraction of the trunk muscles for the development of their endurance and strength characteristics'.

How to perform a Pallof Hold

This can be performed with a resistance band, cable or even a rope with training partner offering the tension.

- Take the rope/cable and hold it at your sternum (arms fully extended).
- Point your left shoulder towards the point to which the rope/cable is attached.
- Stand far away enough to feel the tension.
- Keep your shoulders and hips square.
- Hold the rope/cable straight out in front of your body.
- Resist the tendency to rotate and hold for 60 seconds.
- Keep your abs tight.

Core Principle 3: Learn to hang

Hanging is a lost art form and forgotten method to condition your core.

This is based on research published by the American Association for Health, Physical Education and Recreation that studied muscle activation in the stomach during 10 strenuous abdominal exercises. What they found was 'Intensity of contraction was greatest in the basket hang, followed by three variations of the hook sit-up.'

It beats the conventional sit-up in terms of muscle activation. Researchers added 'The apparently strenuous nature of the basket hang, which is primarily a movement of thigh rather than trunk flexion, implies that this exercise may be useful in the abdominal training of highly conditioned athletes.'

How to perform a twisted hanging leg raise

This is a more difficult progression of the L-sit.

- Lift your legs towards one shoulder and go as high as you can.
- You'll need to tilt your pelvis slightly forward to complete the movement.
- Hold for a second at the top, then slowly lower your legs.
- Repeat on other side.

Core Principle 4: Learn to control

This core principle is linked to an ab exercise launched into the mainstream consciousness after featuring in the Rocky IV training montage set in rural Russia. Known as 'The Dragon Flag', this exercise – unlike most conventional core conditioning – forces the muscles of the stomach to eccentrically contract. They are in tension, but lengthening. This is much like the downward phase of a bicep curl.

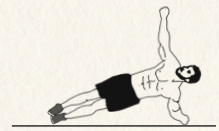
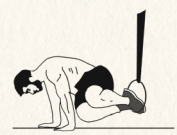
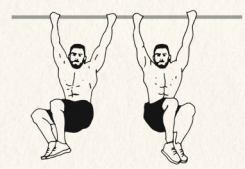
Why is this important? Because research from *The Journal of Applied Physiology* concluded that 'eccentric exercise offers a promising training modality to enhance performance'. Put more simply, forcing the muscles to contract and work during the downward phase is often forgotten. However, it can provide a valuable training stimulus to improve the strength and functionality of any muscle group.

In this instance, your newly chiselled, Rocky-inspired six-pack.

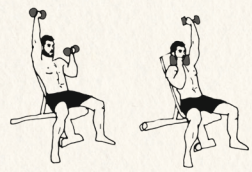
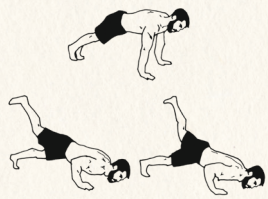
How to dragon-flag

- Lie on the floor while holding onto something stable with your hands by your head.
- With only your head and shoulders in contact with the floor, raise the rest of your body off the floor.
- Keeping as straight as possible, lower yourself to the floor.
- Pause for one second when at the bottom of the exercise.
- Then return to the upright position, all the time ensuring only your head and shoulders are in contact with the floor.

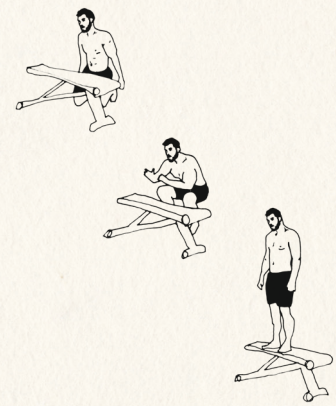
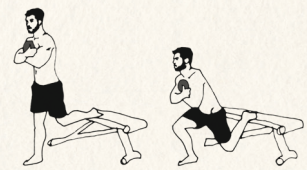
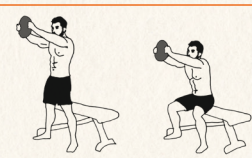
5-DAY ROUTINE | WEEKS 5–8 | SATURDAY: BEGINNER AB WORKOUT

Exercise	Reps	Sets	Rest	Coaching Cues	
T-Plank	10 (each side)	3	90 secs	Start by getting into a conventional plank position. Bend the elbows. Rest your weight on your forearms. Your body should form a straight line, shoulders to ankles. Engage your glutes and abs. Rotate your body so your weight is on the side of one foot. Raise an arm and form a T shape with it. Hold this position, lower your arm and repeat on the other side.	
Suspended Side Crunch	10 (each side)	3	90 secs	Suspend your feet in TRX equipment or gymnastic rings. Draw your attention to your core and focus on maintaining a clean, straight plank, belly button locked in. Draw both knees towards the outside of one elbow, maintaining straight arms and a tight core. Maintain your focus on your tight core and return both legs to a straight start position. Repeat with both legs to the other elbow and return to a start position to complete 2 reps.	
Pallof Hold	60 seconds (each side)	3	60 secs	This can be performed with a resistance band, cable or even a rope with training partner offering the tension. Take the rope/ cable and hold it at your sternum (arms fully extended). Point your left shoulder towards the fixed point. Stand far away enough to feel the tension. Keep your shoulders and hips square. Hold the rope/cable straight out in front of your body. Resist the tendency to rotate and hold for 60 seconds. Keep your abs tight.	
Basket Hang	10 (each side)	3	60 secs	Hang from a pull-up bar. Place the hands shoulder-width apart. Contract your lower abs. Bring your knees to your chest. Twist your hips to one side in a controlled manner, keeping your chest forward at all times. As you do crunch your ribs to your hips. Pause, then perform on the other side.	
Double-knee Tuck Dragon Flag	10	3	60 secs	Lie on the floor while holding onto something stable with your hands by your head. With only your head and shoulders in contact with the floor, raise your entire body from the floor. Keeping as straight as possible, but tucking the legs, lower yourself to the ground. Pause for 1 second when at the bottom of the exercise, then return to an upright position, all the time ensuring only your head and shoulders are in contact with the floor.	
Bear Crunch	60 secs	3	60 secs	From a plank position, bring your right knee under and across your body to meet your left elbow. That is one rep. Now repeat to the other side. When done correctly this move challenges your core stabilisers and your obliques simultaneously.	

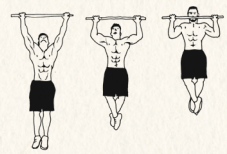
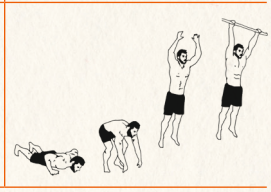
6-DAY ROUTINE | WEEKS 9–12 | MONDAY: 'PUSH' TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Close-grip Plyometric Push-up	5	3	90 secs	Get in a press-up position, hands shoulder-width apart and back straight. Lower your chest to the floor, then push up explosively until your arms fully extend. Attempt to add claps into the movement while airborne if possible. (If you can't manage full press-ups, do them on your knees.)	
One-arm Push-up	10 (each arm)	3	90 secs	Get into push-up position, with one hand on the surface and spread your feet wide apart. Tense your entire body and hold your free hand tight against your lower back. Lower your body slowly until your chest nearly touches the floor, then explode up to the starting position. (If you can't manage full press-ups, do them on your knees.)	
Alternate Arnold Press	12 (each arm)	3	60 secs	Grab a set of dumbbells and bring them to shoulder height, with arms bent and palms facing the body. Gripping the dumbbells as tightly as possible, press one overhead (keeping the other dumbbell at shoulder height) rotating the dumbbells until the palms are facing forward at the top of the press. Pause, and then slowly lower the weight, reversing the rotating so that the palms end facing the body at the bottom of the press. Repeat on the over arm.	
Uchi Mata Push-up	6 (each leg)	3	60 secs	A push-up inspired by a judo throw that increases activation of your core, lower back and hamstrings and boosts demand on the shoulder muscles. Starting in a push-up position, lift your right foot so your leg is parallel to the floor. Lower your body until your chest nearly touches the floor. Raise your lifted leg higher into the air. Push back to the starting position. Switch legs and repeat.	
Incline Dumbbell Fly into Hex Press	12	3	60 secs	Lie flat on the bench holding 2 dumbbells above your chest with your palms facing each other. With a slight bend at your elbows, lower the dumbbells down to your sides until you feel your chest stretch. Bring the dumbbells into the centre of your chest and push them back up to the start position and repeat.	
Weighted Bear Crawl	10 m forward then 10 m backward	3	45 secs	A tough final test to finish the last pushing session of the 12-week programme, here we add weights to what is an already hard exercise in the bear crawl. Complete over 10 m forward and backward and (optional) on the final set, go until complete failure.	

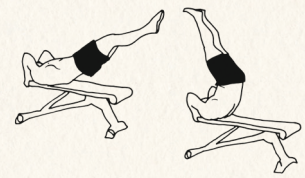
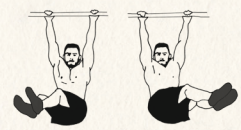
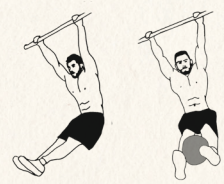
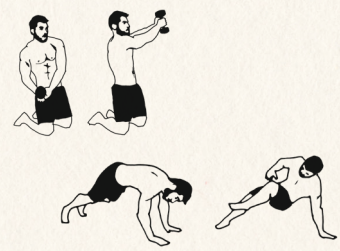
6-DAY ROUTINE | WEEKS 9–12 | TUESDAY: LOWER-BODY TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Knee-to-feet Box Jump	5	3	90 secs	Knee jumps will develop the posterior chain (lower back, glutes and hamstrings) and by adding a box jump after it becomes a continuous movement where you are jumping twice in a row without any time to rest. To perform them, get down on your knees with the top of your feet flat on the ground behind you. Sit back on your feet, lean forward, and bring your arms behind you. Swing your arms forwards, while simultaneously extending your hips forwards. As you come up off the ground, raise your knees up and plant your feet on the ground to catch yourself. Try to land as softly as possible and immediately go into a box jump.	
Bulgarian Split Squat with Plate	12 (each leg)	3	60 secs	Stand facing away from the bench, holding a plate against your chest. Rest one leg on the bench behind you (the laces of your shoes facing down). Squat with your standing leg until the knee of your trailing leg almost touches the floor. Push up through the front foot to return to the start position.	
Paused Goblet Squat	12	3	60 secs	Standing tall with your feet shoulder-width apart, grab the head of a dumbbell with both hands and hold it vertically in front of your chest. Keeping your back straight, squat down until the crease of your hip drops below the knee and the tops of your thighs are at least parallel to the floor. Pause and hold for 3 seconds. Extend your hips and knees to return to the starting position.	
Plate Extended Squat	12	5	60 secs	A conventional squat while holding a weighted plate with your arms extended in front of your chest to increase activation of the core. Lower into a squat position and ensure your thighs reach slightly below parallel.	
Forward and Reverse Lunge into **Forward and Reverse Duck Walk**	10m (forward and back)	4	60 secs	A brutal way to finish any leg session, here we combine the finishing exercises of the previous lower-body workouts and perform them back-to-back in what will be a gruelling leg-specific fitness challenge.	

6-DAY ROUTINE | WEEKS 9–12 | WEDNESDAY: 'PULL' TRAINING

Exercise	Reps	Sets	Rest	Coaching Cues	
Alternate Dumbbell Sumo High Pull	6 (each arm)	3	90 secs	Stand with your legs wider than shoulder-width apart with a dumbbell in one hand. Lower your body into a squat position until your thighs are parallel to the floor. Push through your heels and raise the dumbbell to your chin, so your arm bends and the elbow is in line with the top of your head. Squat down, change hands and repeat.	
Dead-hang Pull-Up	5	3	90 secs	Raise your feet off the floor and hang so the arms are fully extended. Pull yourself up until your chin is over the bar and repeat, ensuring you go back to the hanging position with arms fully extended so the muscles have to work harder to overcome the loss of momentum.	
Incline Bench Dumbbell Reverse Fly	12	3	60 secs	To begin, lie down on an incline bench with the dumbbells in each hand, palms facing each other (neutral grip). Extend the arms in front of you, perpendicular to the angle of the bench. Keep your legs still and apply pressure with the ball of your toes. This is the starting position. Maintaining the slight bend of the elbows, move the weights out and away from each other (to the side) in an arc motion while exhaling. Tip: Try to squeeze your shoulder blades together to get the best results from this exercise. The arms should be elevated until they are parallel to the floor. Slowly lower the weights back down to the starting position while inhaling.	
Upright Row	20	3	60 secs	Hold a dumbbell in each hand, resting in front of your thighs. Lift the dumbbells vertically until they're in line with your collar bone with your elbows pointing out. Lower the dumbbells back down and repeat.	
Spider Zotterman Curl	10	4	60 secs	Lie down on an incline bench. Hold a dumbbell in each hand at arm's length, elbows close to the torso. and palms facing each other. This is the starting position. Holding the upper arm stationary, curl the weights while contracting the biceps as you breathe out. Only the forearms should move. Your wrist should rotate so that you have a supinated (palms-up) grip. Continue the movement until your biceps are fully contracted and the dumbbells are at shoulder level. Hold for a second as you squeeze the biceps. Now, rotate your wrist into a pronated (palms-down) grip with the thumb higher than the pinky. Slowly begin to bring the dumbbells back down. As you reach the lower part of the movement, start rotating the wrist back to a neutral (palms facing your body) grip. Repeat.	
Burpee into Pull-up	100	1	As much as you need to finish 100	The burpee is a four-point move. From standing, drop into a squat with your hands on the ground just in front of your feet. Then kick your feet back behind you, keeping your arms extended so you are in a raised plank position. Complete a push-up, then jump your feet back towards your hands. Squat, then jump into the air with your arms straight above you. Grab the bar and complete 1 pull-up. That's 1 repetition.	

6-DAY ROUTINE | WEEKS 9–12 | THURSDAY: ADVANCED AB WORKOUT

Exercise	Reps	Sets	Rest	Coaching Cues	
Full Dragon Flag	10	4	60 secs	Lie on the floor while holding onto something stable with your hands by your head. With only your head and shoulders in contact with the floor, raise your entire body from the floor. Keeping as straight as possible, lower yourself to the ground. Pause for 1 second when at the bottom of the exercise, then return to an upright position. All the time ensuring only your head and shoulders are in contact with the floor.	
Window Wiper	10 (each side)	3	60 secs	Essentially a basket hang but with straighter legs to add difficulty. Hang from a pull-up bar, hands shoulder-width apart. Contract your lower abs. Bring your knees to your chest. Twist your hips to one side in a controlled manner, keeping your chest forward at all times. As you do, crunch your ribs to your hips. Pause, then perform on the other side.	
Suspended Pike	12	3	90 secs	Get in push-up position. Rest the tops of your feet in the TRX equipment. Keeping your legs as straight as possible, bend your hips and pull your feet towards your chest. Pull your feet towards your hasnds. Hold at the top for three to four seconds, then slowly roll back to the starting position.	
L-sit	60 secs	3	60 secs	Hang from a bar or rings. Place your hands about shoulder-width apart. Keep your legs straight and together. Using your lower abs and hip flexors, lift your legs. Once parallel to the floor hold. Hold your legs at a 90° angle to your torso. This is the L-sit position. Once you've mastered this, try adding weight.	
Dumbbell Wood-chop into Bear Crunch	10 (each side) into 60 seconds of bear crunches	3	60 secs	Hold your dumbbell with both hands next to the outside of your right thigh. Twist your torso to the side and lift the weight up and across your body with straight arms. As you lift, turn your torso so you end up facing the dumbbell, which is above your left shoulder. You should be using your core muscles to control the movement. Return to the starting position, reversing the twist and bringing the weight down as if chopping wood.	

6-DAY ROUTINE | WEEKS 9–12 | FRIDAY: FULL-BODY CAVEMAN CONDITIONING

Congratulations you've made it to weeks 9–12.

Welcome to the bigger brother of The Mountain, which we call The Everest! This takes Caveman Conditioning to a new level as you aim to complete 44.24 seconds of mountain climbers at varying angles, resting for 44.24 seconds only after finishing your hands-elevated mountain climbers.

Repeat this process 10 times.

Why 44.24 seconds?

Because it's double the time you complete for The Mountain, plus 1 set (with rest) equals 176.96 seconds, symbolic of the height of Everest if you climbed up and down it (17,696 m).So your goal is to conquer this entire workout in 1769.6 seconds (or 29.49 minutes)

Exercise	Reps	Sets	Rest	Coaching Cues
Feet Elevated Mountain Climber	44.24 secs	10	44.24 secs after the last set	Set up on the floor as though in a sprinter's blocks, with your feet resting on a bench. Explosively piston your knees in towards your chest. Repeat for 44.24 seconds. Go straight into the next move without rest.
Mountain Climber	44.24 secs			Set up on the floor as though in a sprinter's blocks. Explosively piston your knees in towards your chest. Repeat for 44.24 seconds. Go straight into the next move without rest.
Hands Elevated Mountain Climber	44.24 secs			Set up on the floor as though in a sprinter's blocks with your hands resting on a bench. Explosively piston your knees in towards your chest. Repeat for 44.24 seconds. Then rest for 44.24 seconds (enjoy it) before starting another round of the three moves. Do 10 rounds.

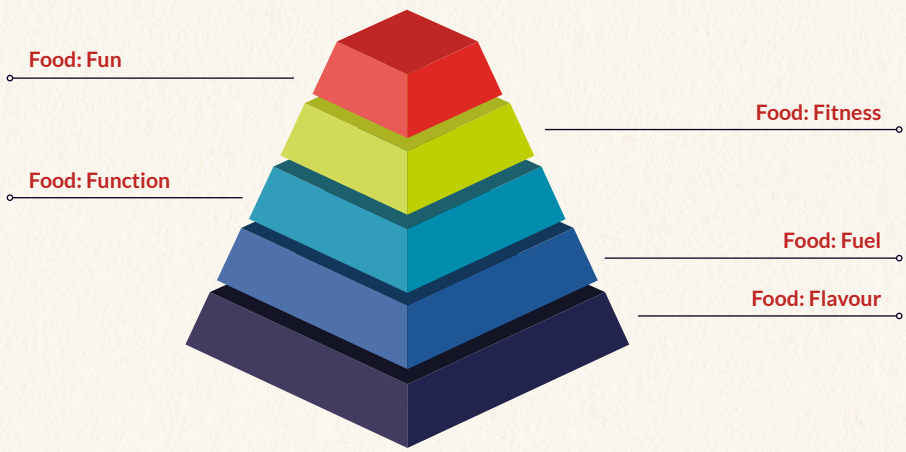
6-DAY ROUTINE | WEEKS 9–12 | SATURDAY: SPORT SPECIFIC

Success! After nine weeks you can now begin to specialise...

You have built greater kinaesthetic awareness and unlocked your neurological potential through the Soviet Union sporting concept of General Physical Preparedness. Increased your work capacity inspired by the bodyweight exercise progressions of the Royal Marines. Now you are ready to perform Directly Associated training based on Geoff's four-point (multi-skill) training system.

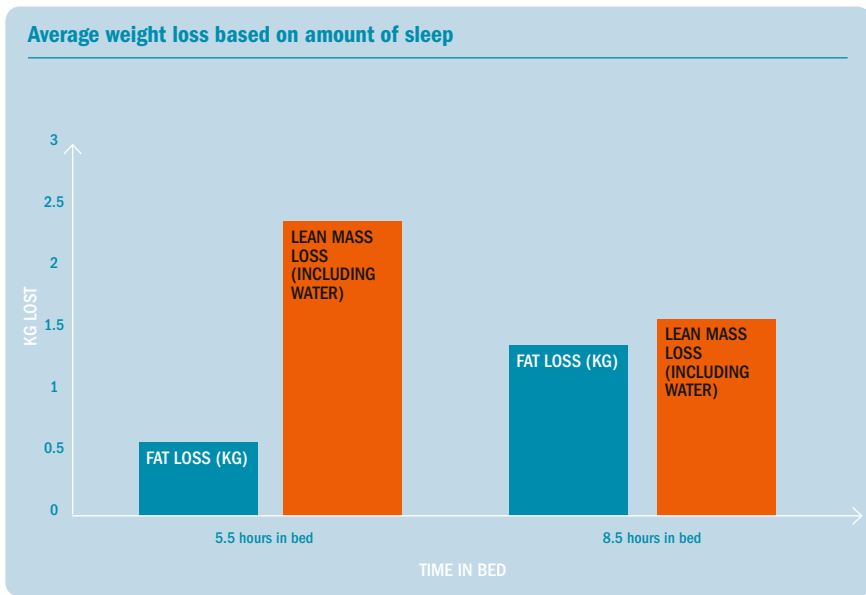
Whether this is football, rugby, running or swimming — or in my case a Tree-athlon — you now 'know your body' better and will be able to learn new movements, with greater proficiency, far faster.

The Fat-loss Pyramid of Priority



PART III: THE SECRETS OF RECORD-BREAKING FITNESS

BODY FAT AND BEDTIME | THE RESULTS



HOW TO WRITE YOUR OWN DIET

IN FIVE STEPS

The best diet in the world is useless unless you can stick to it, so be sure to get stuck into the meal plans and recipe section to make sure your taste buds love your diet plan as much as your waistline does.

1. Work out your basal metabolic rate (BMR)

This is the amount of calories your burn per day at rest.

To do this, use the Harrison Benedict Formula.

- **For men:** 66.5 + (13.75 x weight in kg) + (5.003 x height in cm) – (6.755 x age in years)
- **For women:** 655.1 + (9.563 x weight in kg) + (1.850 x height in cm) – (4.676 x age in years)

2. Multiply by your level of activity

- Little to no exercise means multiply your BMR by 1.2.
- Light exercise a few times a week by 1.375.
- Moderate exercise 3–5 times a week by 1.55.
- Heavy exercise 6–7 times a week by 1.725.

Congratulations! You now know your daily calorie requirements.

3. Set your calorie deficit/surplus

Determine whether you want to…

- Eat more calories than you burn to store fat and set the size of your calorie surplus.
- Burn more calories than you eat to lose fat and set the size of your calorie deficit.
- Eat the same as you burn to stay the same (calorie balance).

4. Find your food ratio

This part is important. To re-quote the *International Journal of Obesity* 'Regardless of assigned diet, 12-month weight change was greater in the most adherent.' Therefore the ratio of protein, fats and carbohydrates doesn't matter as long as you can stick to it. To help

you find your food ratio, here are three to choose from:

- High Carb: 20% protein, 10% fats, 70% carbohydrates
- Balanced Macros: 25% protein, 25% fats, 50% carbohydrates
- High Fat: 23% protein, 70% fats, 7% carbohydrates

5. Make your meals in grams

Now it's time to move your food from the calculator and into the actual kitchen. Protein and carbohydrates contain 4 calories per gram, while fat provides 9 calories per gram. Based on this, here's how a 2000-calorie diet looks for each food ratio:

HIGH CARB	BALANCED MACROS	HIGH FAT
Protein = (2000 x 0.2) ÷ 4 = 100g	**Protein** = (2000 x 0.25) ÷ 4 = 125g	**Protein** = (2000 x 0.23) ÷ 4 = 115g
Fats = (2000 x 0.1) ÷ 9 = 22g	**Fats** = (2000 x 0.25) ÷ 9 = 56g	**Fats** = (2000 x 0.70) ÷ 9 = 156g
Carbs = (2000 x 0.7) ÷ 4 = 350g	**Carbs** = (2000 x 0.50) ÷ 4 = 250g	**Carbs** = (2000 x 0.07) ÷ 4 = 35g

1 MAN | 3 DIETS | 1 RESULT

Age: 27 | Height: 178cm (5ft 10) | Weight: 90kg | Total calories per day: 3823

HIGH CARB			BALANCED MACROS			HIGH FAT		
20% Protein	10% Fats	70% Carbs	20% Protein	25% Fats	55% Carbs	20% Protein	70% Fats	10% Carbs
765 calories of protein	382 calories of fats	2676 calories of carbs	765 calories of protein	956 calories of fats	2103 calories of carbs	765 calories of protein	2676 calories of fats	382 calories of carbs
191g of protein a day	42g of fats a day	669g of carbs a day	191g of protein a day	106g of fats a day	526g of carbs a day	191g of protein a day	297g of fats a day	96g of carbs a day

Inspired by the *Journal of Sports Medicine*[72]

Inspired by research from the *Journal of the American Dietetic Association*[73]

Inspired by the *European Journal of Applied Physiology & Occupational Physiology*[74]

Serving as a human guinea pig for three different diets

7-DAY HIGH CARB MEAL PLAN

HIGH CARB		
20% Protein	10% Fats	70% Carbs

	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6	DAY 7
Breakfast	Quinoa cooked in coconut milk topped with fruit and almonds	Strawberry and pineapple protein pancakes	Oats/Porridge, with cinnamon, blueberries with a scoop of whey protein	Nourishing nut and seed bread (see below)	Protein waffles and honey	Granola, yogurt and mixed berries with a scoop of whey protein	Chia breakfast bowl
Lunch	Harissa-crusted chicken	Loaded potato skins (with chilli) and salad	Baked egg, chickpeas, spinach, aubergines and peppers with tortilla wraps	Salmon, cream cheese and toasted bagel	Spaghetti and meatballs	Wholewheat tortilla with shredded chicken and avocado slices	Lamb tagine and rice (see below)
Snack	Healthy biscotti and green tea	2 rice cakes and cottage cheese	Wholegrain protein muffins (see below)	Tortilla chips with chill and salsa dip	Greek yogurt with blueberries, honey and a scoop of whey protein	Sliced banana bread	Protein granola bar
Dinner	Steak burger	Grilled salmon noodle stir-fry (see below)	Baked sweet potato with cottage cheese, salsa and bacon croutons	Seafood linguine	Chicken curry and rice	Chicken, noodles and stir-fry mixed vegetables	Chilli, brown rice and mixed vegetables
Snack	Healthy protein flapjack	Natural yogurt with apple slices and granola	Oatmeal protein cookie with milk	Quinoa cooked in coconut milk covered in fruit and almonds	Lemon and quark granola and oat cheesecake	Warm bowl of porridge oats, cinnamon, raisins and a scoop of apple and cinnamon protein	Healthy protein flapjack

CHIA BREAKFAST BOWL

by Annie Mayne

7-day High Carb Best Meal

Made from one of the oldest superfoods in the world, chia seeds are an incredibly versatile ingredient and can be blended with any fruit to make a breakfast bowl packed with nutrient-dense carbohydrates to start your day.

Ingredients

Serves 1

250ml coconut milk

30g strawberries, blueberries or other fresh fruit

2 tbsp agave syrup

½ tsp vanilla extract

32g chia seeds

Method

1. Add the coconut milk, berries, agave syrup and vanilla to a blender. Blend on high for 1–2 minutes to purée the fruit.
2. Pour the mixture into a mason jar or a small bowl/cup and mix in the chia seeds.
3. Cover and refrigerate for at least 3 hours or preferably overnight.

	Calories	Carbs	Fat	Protein
Total	492	66g	22g	10g
Per serving	492	66g	22g	10g

7-DAY BALANCED MACRO MEAL PLAN

BALANCED MACROS		
20% Protein	25% Fats	55% Carbs

	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6	DAY 7
Breakfast	Scrambled eggs on toast	Blueberry and orange protein pancakes (see below)	Cinnamon and raisin oatmeal with a scoop of whey protein	Raspberry and blueberry protein waffles	Spinach and onion omelette	Banana and honey protein granola	Banana and coconut protein waffles
Lunch	Beef chilli stir-fry	Kidney bean chilli	Parma ham and cheese toasties (see below)	Harissa-crusted chicken with coconut rice and stir-fried greens	Beef kofta with bulgur and kale salad	Quinoa toast with peanut butter, avocado and strawberry jam spreads	Chicken burrito with avocado and spinach
Snack	Raw protein balls	Protein granola bar	Pumpkin muffins	Super green protein smoothie (see below)	Club sandwich	Protein smoothie with pineapple and Raspberry	Raw vegan brownies with a protein shake
Dinner	Roasted chicken thighs, potatoes and spring onions with a herb vinaigrette	Steak and mixed vegetables (see below)	Chicken and vegetable flat-crust pizza	Cauliflower and aubergine pizza	Steak and sweet potato fries	Beef and cheese burger	Pork, beef, pancetta meatballs
Snack	Mixed berry smoothie	Turmeric chicken salad	Chocolate acai protein bowl	Fruit and nut mix and a protein shake	Protein pumpkin pie muffins	Low-fat Greek yogurt and sliced apple	Blueberry, strawberry and pear smoothie

HARISSA-CRUSTED CHICKEN WITH COCONUT RICE AND STIR-FRIED GREENS

by Simon Partridge

7-day Balanced Macro Best Meal

The beauty of this recipe is it's so easy to make, tastes amazing and because of the fresh ingredients used, it has the perfect combination of protein, fats and carbohydrates on every plate.

Ingredients

Serves 1

2 skin-on chicken breasts
2 tsp harissa
Salt and freshly ground pepper
1 tbsp coconut oil
100g long grain rice
1 tin coconut milk
6 spears asparagus, cut lengthways
1 pak choi, leaves separated
50g mangetout
1 courgette, thinly sliced
1 handful baby spinach
50g fresh coriander, chopped
Juice of 1 lime

Method

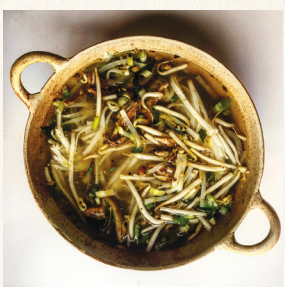
1. Preheat the oven to 200°C.
2. Smear the harissa on to the skin side of the chicken.
3. Season with salt and black pepper and drizzle a little coconut oil over the top.
4. Roast for 20–25 minutes. In the meantime, cook the rice. Measure your rice out in cups: if your 100g dried rice is equal to 1 cup, then add 1 cup of boiling water to it and then 1 cup of coconut milk. Add a pinch of salt and simmer for 12–15 minutes until almost cooked, then remove from the heat. Cover with a lid. Adding the coconut milk and then leaving it to rest will produce the perfect sticky rice.
5. While the chicken is cooling and the rice is resting, heat the remaining coconut oil in a wok, then add the vegetables and stir-fry for 2 minutes.
6. Add the coriander and lime juice.
7. Serve immediately. It's very simple, very quick and very tasty!

	Calories	Carbs	Fat	Protein
Total	935	53g	60g	48g
Per serving	935	53g	60g	48g

7-DAY HIGH FAT MEAL PLAN

HIGH FAT		
20% Protein	70% Fats	10% Carbs

	DAY 1	DAY 2	DAY 3	DAY 4	DAY 5	DAY 6	DAY 7
Breakfast	Avocado and scrambled eggs with crusty roll cooked with coconut oil	Full-fat Greek yogurt mixed with peanut butter	Smoked salmon, cream cheese and poached eggs	Full-fat English breakfast with avocadoes	Baked eggs with bacon and creamy kale	Simple omelette with spinach	Smoked salmon, avocado and poached egg
Lunch	Roasted veg and feta cheese salad (see below)	Chorizo, peppers and baked egg	Chicken skewers, peanut satay sauce and feta cheese salad	Loaded hasselback zucchini (with cheese, bacon and spring onions)	Chicken burger and veg	Chicken, cashew and spring onion stir-fry	Hummus, falafel and feta cheese salad
Snack	Dark chocolate peanut butter cups	Celery sticks dipped in almond butter	Vegan protein balls	Protein smoothie blended with almond butter	Hummus and carrot/celery sticks	Chicken pho soup (see below)	Green tea/ Coffee/Lady Grey tea mixed with a teaspoon of coconut oil
Dinner	Grilled seabass	Boiled egg, avocado and chickpea salad (see below)	Chicken meatballs with courgette and spinach	Grilled steak burger, veg and cashews	Fillet steak with sweet potato fries and greens	Venison burger with bacon, melted cheese and greens (see below)	Roasted butternut squash, goat's cheese, spinach and olive quiche
Snack	Chocolate and almond milk protein smoothie	Dark chocolate peanut butter cups	Almond and avocado smoothie	Smoked salmon, and poached egg	Coconut milk protein smoothie	Keto-friendly vegan chocolate tart	Simple omelette with spinach and avocado



DARK CHOCOLATE PEANUT BUTTER CUPS

by Annie Mayne

7-day High Fat Best Meal

Simply the greatest snack you can have on a high-fat diet. Packed with healthy fats, these are practically a dessert and have so much energy crammed into every cup.

Ingredients

Makes 6

125g dark chocolate

2 tbsp (natural) peanut butter

2 tsp organic virgin coconut oil from THE PROTEIN WORKS™ (if this isn't your jam, substitute it with grass-fed butter instead. It has the same healthy fat which solidifies the peanut butter middles)

A small pinch of sea salt

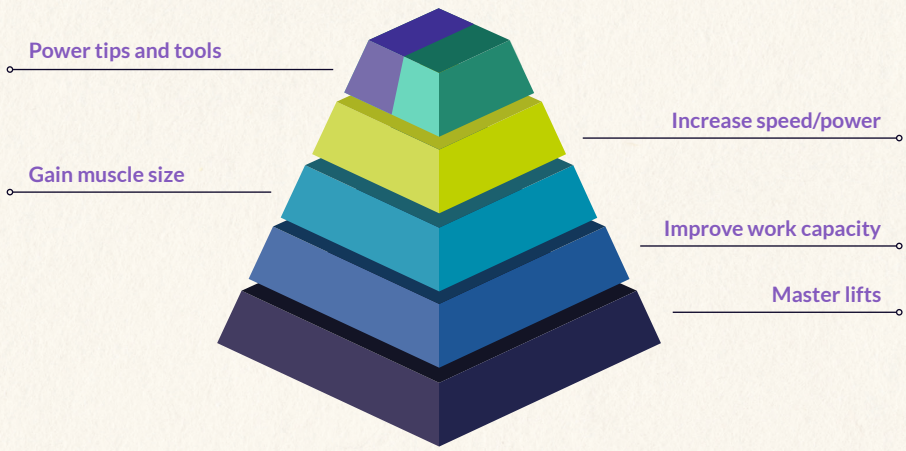
6 walnut halves

Method

1. Melt the chocolate, either in the microwave or in a heatproof bowl over a pan of gently steaming water.
2. Stir until smooth, then use a spoon to drop a small amount of the chocolate into the bottom of 6 cupcake cases. (To ensure you don't have spillage, I recommend double layering the cases).
3. With the same spoon, push the chocolate up the sides of the case to create a cup. This will hold the mixture.
4. Place in the freezer for about 30 minutes.
5. In the meantime combine the peanut butter, coconut oil and salt. Stir until you get a smooth and creamy consistency.
6. Remove the chocolate cups from the freezer and add a dollop of peanut butter in the middle of each. If you can, try to leave some room around the sides so that the chocolate can cover the sides when filling. I have yet to master that, as demonstrated by my photo!
7. Cover each cup with a layer of melted chocolate, decorate with a walnut half and return to the freezer for about 2 hours, or longer if needed.
8. Remove the paper cups once set. Store in the freezer or fridge for best consistency.

	Calories	Carbs	Fat	Protein
Total	928	52g	91g	18g
Per serving	155	9g	15g	3g

The Strength, Speed and Power Pyramid of Priority



THE SQUAT

'The squat is a great lower-body developer, but – as you just found out – it's really a total body exercise, working every muscle you have, albeit with an emphasis on the legs.'

ANDY BOLTON

People are squatting wrong. According to Andy 'Too many complain about back pain, knee pain and even shoulder and elbow pain when squatting. I'd say 95% are doing it wrong.

But if people spent more time learning the right technique, as opposed to just mindlessly pumping out the reps, they could squat pain free. Improving strength and reducing injury at the same time.'

Enter Andy's Three-point Squat Plan:

- How to set up.
- How to lower the weight.
- How to lift the weight.

How to set up

When the bar is in the rack position, it should be in line with your mid-chest.

Too high and you have to calf-raise it off the rack, too low and you have to do a quarter squat to unrack the bar. Neither option works safely, especially as you get stronger and the weights get heavier.

Grip the bar with the narrowest grip you can that doesn't cause elbow pain (you'll have to experiment with this).

Step under the bar with a hip-width stance, placing your feet directly under the bar. Your shoulders should be pulled back and down, your chest should be out, your upper back tight and your lower back in a neutral position.

Take a deep breath of air, get as tight as you can and unrack the bar.

To unrack it, simply stand straight up with it. Now step back 20–25 cm (4–6 inches) with your left foot and then do the same with your right foot.

Your stance should be shoulder width or slightly wider, your toes turned out somewhere between 5 and 35 degrees depending on your flexibility.

You are now ready to squat.

How to lower the weight

Pause. Take some air in. Get as tight as possible.

Push your knees out slightly and 'sit back' as you descend.

Aim to keep your shins as vertical as possible. They will shift forwards somewhat, but should feel like they stay pretty vertical.

Under no circumstances should your heels leave the floor.

When the crease of your hip is just below the top of your knee (when viewed from the side) you have hit the bottom of the squat… you are 'in the hole'.

How to lift the weight

Do not breathe out when you're in the hole. This will cause you to lose tightness and therefore you'll lose strength and greatly increase your injury risk. Keep the abs as tight as possible and squeeze the bar hard.

As you come up, avoid letting your knees cave in and carry on looking straight ahead or slightly up.

Keep your upper back tight with your shoulders back and down throughout.

When you reach lockout, stay tight.

Take in more air and either step forwards and rack the bar or descend into another rep.

On the descent, move as quickly as you can while maintaining complete control of the bar (Note: This may not actually be very quick at all.)

On the ascent, drive the bar upwards as hard and as fast as you possibly can. Think speed, speed, speed. That said, never compromise technique for speed. Nail your technique first and then add speed in once you become a competent and efficient squatter.

THE BENCH PRESS

'Every man (and some women) wants a bigger bench. Who hasn't been asked the famous question: How much do you bench?'

ANDY BOLTON

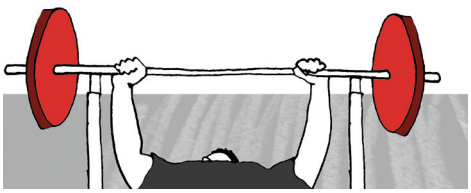
The first step towards building a massive bench press is (again) technique.

Andy says 'The way most bodybuilders (even the pros) and most gym rats bench is a disaster. The typical 'flat back, flared elbows, touching the upper chest' style of benching is a nightmare for the shoulders. It beats them up and will lead to injury sooner or later. I've seen it happen too many times. Not only that, it's weak, too.'

He adds, 'There's only one way a self-respecting human being should bench and that's powerlifting style. Don't worry, it'll build just as much muscle as the style that destroys your shoulders, but it'll also let you lift way more weight and massively reduce the risk of injury.'

Enter Andy's Three-point Bench Press Plan…

- How to set up.
- How to lower the weight.
- How to lift the weight.



How to set up

Lie on the bench so your eyes are directly under the bar.

Grip the bar with your hands slightly wider than shoulder-width apart. If you have a history of shoulder pain or injuries, take a narrower grip – index fingers 2.5 cm (1 in) from the smooth part of the bar.

Spread your feet as wide as you can, drive your knees out and tense your glutes as hard as possible. This creates a stable base. (Think in terms of a pyramid – they didn't build them with a narrow base so you shouldn't bench with a narrow stance either.)

Pinch your shoulder blades together and force your chest out to create a slight arch in your back. (The more you arch the shorter the range of motion, but the greater the risk of back injury.) However high you decide to arch, your shoulder blades must be pulled back and down and your chest must be forced out.

(No 'flat back' benching allowed here – it's rough on the shoulders.)

Take a deep breath into your belly, and with the help of a spotter if necessary, contract your lats hard and pull the bar out of the rack. Pull it out until it's directly over your nipple line.

You're now in the start position and ready to take the bar for a ride.

How to lower the weight

While you push your feet into the floor, drive your knees out and keep your glutes as tight as possible.

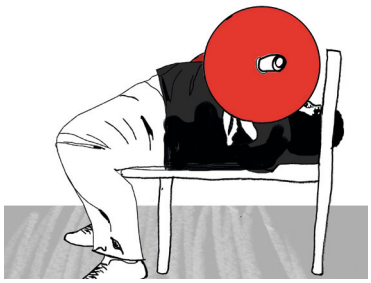
Bring the bar down in a straight line so that it touches your nipple line.

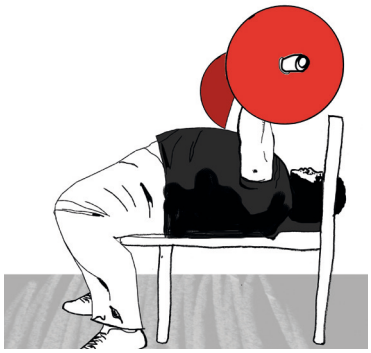
As you descend, grip the bar, keep your lats tight and ensure your shoulders stay back and down. Do not bounce the bar off your chest.

Maintain control at all times.

How to lift the weight

As the bar touches your chest, ensure everything is tight.





Holding your breath, immediately drive the bar off your chest and towards lockout.

Push the bar in a straight line or slightly towards your face. Do not push it way back towards your face, as this can be rough on the shoulders.

If you want, breathe out at the sticking point – while keeping tight. Otherwise, breathe out a little at the top.

Take in some more air and descend into another rep.

THE DEADLIFT

'The deadlift is a brutal test of strength. There's a bar on the floor; your mission is simply to grab it with straight arms and then stand up straight with it in your hands.'

ANDY BOLTON

In theory the deadlift sounds pretty simple, but the first thing I learnt from Andy was that when pulling big weights, the reality is very different. 'There's very little margin for error. When shifting a lot of poundage you can injure your back and pop a bicep faster than you can imagine,' he says. 'But learn the correct technique and those dangers can be more or less eliminated.'

Enter Andy's Three-point Deadlift Plan:

- How to set up.
- How to lift the weight.
- How to lower the weight.

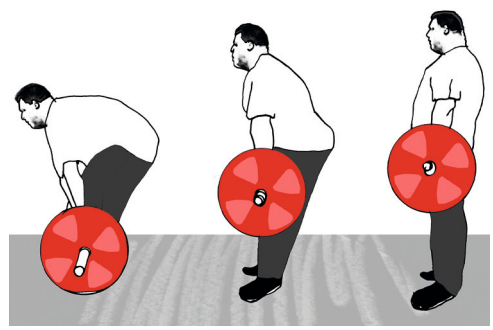
How to set up

Approach the bar with a hip-width stance.

Point your toes straight ahead or turn them out very slightly.

Be sure to deadlift wearing lifting slippers or go barefoot.

Relax your upper back. Keep your lower back neutral and hinge at the hips. Then (with little knee bend) lower yourself to the bar.



Take a deep breath of air into your belly and pull yourself down to the bar until it's touching your shins and your hamstrings feel tight and loaded. Your upper back can be rounded, but your lower back must still be in a safe, neutral position.

At this point you have two choices:

Grip Option 1: Use the Olympic lifting hook grip. The hook grip is double overhand, but the index and middle fingers cover the thumb. This is pretty uncomfortable at first, but you'll get used to it.

Or… **Grip Option 2:** Use the powerlifting style mixed grip where one hand is in the same overhand position as the double overhand grip, but the other hand is under the bar, palm facing backwards. If you use this mixed grip, be sure to switch hand placement on every set to avoid uneven upper back development.

How to lift the weight

Now, with a firm grip, drive the bar off the floor as hard as you can, keeping the bar as close to your body as possible.

When the bar gets just past your knees, drive your glutes forwards to lock the bar out. Stand perfectly erect at lockout – do not lean back.

How to lower the weight

Take a sip of air and descend quickly.

Alternatively, drop the bar, reset your stance and grip, and pull another rep.

Calculate your One-Rep Max (1RM)

Calculate your one-rep max (1RM)	
Repetitions	**% one-rep max (1RM)**
1	100
2	95
3	93
4	90
5	87
6	85
7	83
8	80
9	77
10	75
11	73
12	70

Increase Strengeth | Avoid Injury

Study	Sport	Injury rate per 1000 hours	Most common regions
Winwood	Strongman	5.50	Low back
Keogh (2006)	Powerlifting	4.40	Shoulder
Quinney (1997)	Powerlifting	3.70	Low back
Calhoun (1999)	Olympic weightlifting	3.30	Low back
Hak (2013)	CrossFit	3.10	Shoulder
Raske and Norlin (2000)	Olympic weightlifting	2.60	Low back
Raske and Norlin (2000)	Powerlifting	2.60	Shoulder
Haykowsky (1999)	Powerlifting	1.10	Low back and shoulder
Siewe (2011)	Powerlifting	1.00	Shoulder
Eberhardt (2007)	Bodybuilding	1.00	Not specified
Brown (1986)	Powerlifting	0.84	Shoulder
Siewe (2014)	Bodybuilding	0.24	Shoulder

YOUR 12-WEEK STRENGTH WORKOUT

	3-DAY ROUTINE	4-DAY ROUTINE	5-DAY ROUTINE
	WEEKS 1–4	**WEEKS 5–8**	**WEEKS 9–12**
KEY FOCUS	Practise technique Increase muscle strength Functional Strength Training Powerlifting-centric	Practise technique Increase work capacity Functional & Structural Improve muscle size & strength	Practise technique Increase muscle size Structural Strength Training Bodybuilding-centric
WORKOUT	**Monday:** Bench Press **Tuesday:** REST **Wednesday:** Squat **Thursday:** REST **Friday:** Deadlift **Saturday:** REST **Sunday:** REST	**Monday:** Bench Press **Tuesday:** Deadlift **Wednesday:** REST **Thursday:** Squat **Friday:** Shoulder Press **Saturday:** REST **Sunday:** REST	**Monday:** Bench Press **Tuesday:** Deadlift **Wednesday:** Arms **Thursday:** Squat **Friday:** Shoulder Press **Saturday:** REST **Sunday:** REST
MAIN GOAL	Refine and improve technique to unlock the neurological potential of the muscle mass you have and increase strength.	Improve work capacity by introducing more repetitions and sets, therefore improving your body's ability to tolerate more strength training.	Emphasise 'bodybuilding-centric' movements to improve the size of your muscles and therefore their 'potential strength'.

3-DAY ROUTINE | WEEKS 1–4 | MONDAY: BENCH PRESS

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Bench Press	5	3	120 secs	87%	Functional Strength
Close Grip Bench	8	3	90 secs	80%	Functional Strength
Dumbbell Fly	12	3	60 secs	70%	Structural Strength
Tricep Pushdown	20	3	60 secs	60%	Structural Strength
Tyre Flip	10	4	60 secs	Load with hard but achievable weight	Work Capacity

3-DAY ROUTINE | WEEKS 1–4 | WEDNESDAY: SQUAT

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Squat	5	3	120 secs	87%	Functional Strength
Leg Press	8	3	90 secs	80%	Functional Strength
Leg Extension	12	3	60 secs	70%	Structural Strength
Calf Raise	20	3	60 secs	60%	Structural Strength
Backwards Sled Drag	20 metres	4	60 secs	Load with hard but achievable weight	Work Capacity

3-DAY ROUTINE | WEEKS 1–4 | FRIDAY: DEADLIFT

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Deadlift	5	3	120 secs	87%	Functional Strength
Pull-up	8	3	90 secs	80%	Functional Strength
Bent-over Row	12	3	60 secs	70%	Structural Strength
Barbell Curl	20	3	60 secs	60%	Structural Strength
Battle Ropes	60	4	60 secs	-	Work Capacity

4-DAY ROUTINE | WEEKS 5–8 | MONDAY: BENCH PRESS

Exercise	Reps	Sets	Rest	% One-Rep Max.	Focus
Bench Press	5	5	120 secs	87%	Functional Strength
Floor Press	8	3	90 secs	80%	Functional Strength
Incline Bench Press	12	3	60 secs	70%	Structural Strength
Ring Dip	12	3	60 secs	70%	Structural Strength
Prowler Push	20 metres	3	60 secs	Load with hard but achievable weight	Work Capacity

4-DAY ROUTINE | WEEKS 5–8 | TUESDAY: DEADLIFT

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Deadlift	5	3	120 secs	87%	Functional Strength
Pull-up	8	3	90 secs	80%	Functional Strength
Bent-over Row (Underhand Grip)	12	3	60 secs	70%	Structural Strength
Barbell Drag Curl	20	3	60 secs	60%	Structural Strength
Kettlebell Swing	20	5	60 secs	60%	Work Capacity

4-DAY ROUTINE | WEEKS 5–8 | THURSDAY: SQUAT

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Squat	5	3	120 secs	87%	Functional Strength
Front Squat	8	3	90 secs	80%	Functional Strength
Calf Raise	20	3	60 secs	60%	Structural Strength
Forward Lunge	20 metres	4	60 secs	Load with hard but achievable weight	Work Capacity
Backward Lunge	20 metres	4	60 secs	Load with hard but achievable weight	Work Capacity

4-DAY ROUTINE | WEEKS 5–8 | FRIDAY: SHOULDER PRESS

Exercise	Reps	Sets	Rest	% One-Rep Max.	Focus
Standing Barbell Press	5	3	120 secs	87%	Functional Strength
Front Dumbbell Shoulder Raise	12	3	90 secs	70%	Structural Strength
Side Dumbbell Shoulder Raise	12	3	90 secs	70%	Structural Strength
Battle Ropes	20	3	60 secs	-	Work Capacity

Building strength with Shaun Stafford, good friend and fitness legend

5-DAY ROUTINE | WEEKS 9–12 | MONDAY: BENCH PRESS

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Bench Press	5	5	120 secs	87%	Functional Strength
Incline Bench Press	12	3	60 secs	70%	Structural Strength
Ring Dip	12	3	60 secs	70%	Structural Strength
Dumbbell Fly	20 – Final set to failure	5	60 secs	60%	Structural Strength
Wide-grip Push-up	20 – Final set to failure	5	60 secs	60%	Structural Strength

5-DAY ROUTINE | WEEKS 9–12 | TUESDAY: DEADLIFT

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Deadlift	5	3	120 secs	87%	Functional Strength
Bent-over Row	8	3	90 secs	80%	Functional Strength
Wide Pull-up (Overhand Grip)	12	3	60 secs	70%	Structural Strength
Farmers Walk	20 metres	5	60 secs	Load with hard but achievable weight	Work Capacity
Shrug	20 – Final set to failure	5	60 secs	60%	Structural Strength

5-DAY ROUTINE | WEEKS 9–12 | WEDNESDAY: ARMS

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Pull-up (Neutral Grip)	5	3	120 secs	87%	Functional Strength
Incline Dumbbell Curl	12	3	60 secs	70%	Structural Strength
Seated Dumbbell Drag Curl	12	3	60 secs	70%	Structural Strength
Weighted Dip	5	3	120 secs	87%	Functional Strength
Tricep Pushdown	20 – Final set to failure	4	60 secs	60%	Structural Strength
Z-bar Triceps Extension	20 – Final set to failure	4	60 secs	60%	Structural Strength

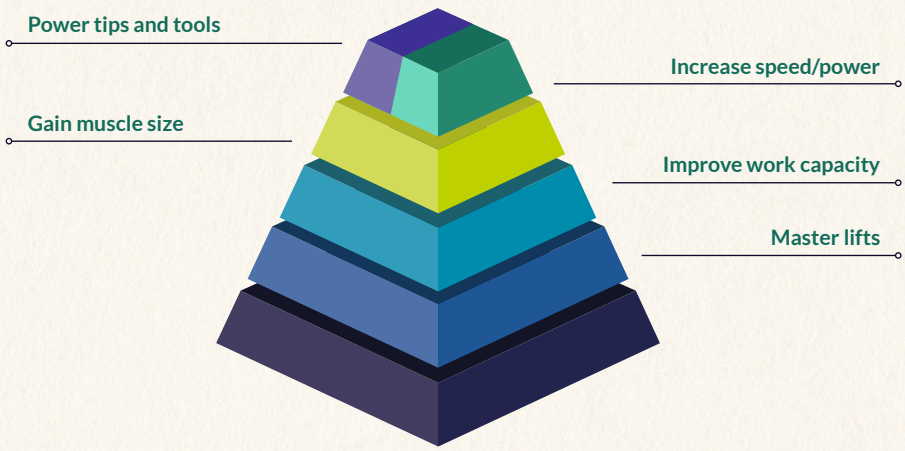
5-DAY ROUTINE | WEEKS 9–12 | THURSDAY: SQUAT

Exercise	Repetitions	Sets	Rest	% One-Rep Max	Focus
Squat	5	3	120 secs	87%	Functional Strength
Leg Press	8	3	90 secs	80%	Functional Strength
Calf Raise	20 – Final set to failure	3	60 secs	60%	Structural Strength
Bulgarian Split Squat	20 – Final set to failure	3	60 secs	60%	Structural Strength
Forward & Backward Lunge	20 metres	4	60 secs	Load with hard but achievable weight	Work Capacity

5-DAY ROUTINE | WEEKS 9–12 | FRIDAY: SHOULDER PRESS

Exercise	Repetitions	Sets	Rest	% One-Rep Max	Focus
Standing Barbell Press	5	3	120 secs	87%	Functional Strength
Front Dumbbell Shoulder Raise	12	3	90 secs	70%	Structural Strength
Side Dumbbell Shoulder Raise	12	3	90 secs	70%	Structural Strength
Face-pull	20 – Final set to failure	3	60 secs	60%	Structural Strength
Hang Power Clean & Press	20 – Final set to failure	3	60 secs	60%	Structural Strength

The Strength, Speed and Power Pyramid of priority



OVERVIEW: OLYMPIC LIFTING

Clean and jerk

The clean and jerk begins with the barbell on the floor. The athlete must then put that weight overhead in two separate motions: clean it to the shoulders (pause) and then jerk (thrust) it overhead so the arms are fully locked out. Believed to be the single best lift for developing strength, speed and power, the entire movement requires a big pull for the clean. A powerful front squat to come out of the squat position. Finally, a herculean jerk to finish with the barbell above your head.

The set-up

- Set your feet hip-width apart and place them directly under the bar.
- Bend over (back straight) and grip the bar with a shoulder-width grip.
- Drop your hips into the squat position.
- Keep the back flat, hips higher than your knees and arms straight.
- Look straight ahead.

The pull

- Start the movement by powerfully pulling the bar with your legs, hips and back.
- Keep the bar close to your shins, but don't make contact with your shins or knees as it rises.
- As it passes your thighs it may brush them making light contact.
- Finish the pull by triple-extending your ankles, knees and hips as you rise onto your toes.

Receiving the bar

- Your body will now be fully extended.
- Immediately shrug your shoulders and ‘pull’ yourself under the bar.
- The goal here is for you to travel as fast as possible below the barbell.
- Move into a full squat as the bar passes your chest as fast as possible.
- Thrust your elbows out in front of you and turn your wrists over.
- Ensure you receive the bar on top of your clavicles and deltoids.

The stand

- Only once you have the bar under control can you begin to stand.
- Do this by driving your heels into the floor.
- Many people complicate this part of the lift, but it’s simply a front squat.
- Note that your torso might lean forward slightly and your hips may shift backward.
- But it’s important that the bar must always stay over your ankles as you stand up.

The jerk

- Now finish the movement by bending your knees and hips about 5–10 cm.
- Next, generate as much force as possible by powerfully extending your knees, hips and elbows.
- Thrust the bar overhead, remembering to lean your head back as the barbell travels above your shoulders.
- As the barbell passes your head, 'split' your legs (one forward and the other back).
- Lock the arms (and bar) overhead as both feet land at the same time.
- Return your head to the neutral position so the bar is directly over your ears (or slightly behind them).
- Finally, bring your feet back together about hip- to shoulder-width apart.
- Stand upright, controlled and motionless.
- That is one repetition.

Snatch

The snatch is unlike the clean and jerk in that it's performed in a single motion. It's considered to be one of the most explosive and powerful movements you can perform in a gym, but don't let Sonny's giant quads fool you. The snatch is not just raw power, but requires finesse, and when it's executed perfectly the heavy barbell should look light as a feather.

Worth pointing out is the perceived ease with which an athlete snatches the bar is a good indicator of their proficiency in Olympic lifts. Remember: strength is a skill and like powerlifting, it's not what you lift but how you lift it.

The set-up

- Start with your feet hip-width apart and position them under the bar.
- Bend over (back straight) and grab the bar with a wide grip.
- This should be approximately 75+ cm but will depend on your height and shoulder flexibility since some athletes will use the entire length of the bar. Get into a squat position. Back flat, arms straight and hips higher than your knees, looking ahead.

The pull

- Begin the movement by lifting with your legs, glutes and back.
- Keep the bar as close to your shins as possible without making contact.
- Avoid the knees, but don't worry if your thighs brush the bar on the way up.
- Finish the pull by triple-extending your ankles, knees and hips as you rise onto your toes.

Receiving the bar

- With your back now extended, shrug your shoulders and 'pull' yourself under the bar.
- Move into a full squat position.
- Once the bar passes your head, turn your wrists over and push the bar to a full lockout.
- Do this by driving with your shoulders and triceps and don't worry if your feet position moves to a shoulder-width jump.
- You're now in a full squat with your torso extended, arms locked and bar directly over (or slightly behind) your ears.

The stand

- Only once you have complete control of the barbell can you begin the stand.
- Extend your knees and hips and ensure the bar always stays directly over your ankles and hips.
- Stand upright, controlled and motionless.
- That is one repetition.

YOUR 12-WEEK SPEED AND POWER WORKOUT

REPETITION AND WEIGHT EXPLAINED

When training for speed the weight, repetitions, rest and sets all change. It's important to note this subtle difference because force and speed have an inverse relationship: as you put more weight on the bar the force goes up, but the speed at which you lift comes down (see the graph opposite).

Maximum Strength Training

Training in this 'zone' means we train heavy! Since, as we know, maximal strength is the amount of force generated through a specific movement, examples include the squat, bench and deadlift at 90–100% of your one-repetition maximum.

Strength–Speed Training

Training in this zone requires an athlete to move less weight than when training for maximum strength, but move it at a faster velocity. Examples include Olympic lifts (snatch, clean and jerk and snatch press) at 80–100% of your 1RM.

Peak Power Training

Training in this zone means we train heavy and fast! Delivering peak power output, these exercises produce the greatest amount of force in the least amount of time, and examples include the pull variations of the clean and snatch, jump squats and medicine ball throws at 30–80% of your 1RM.

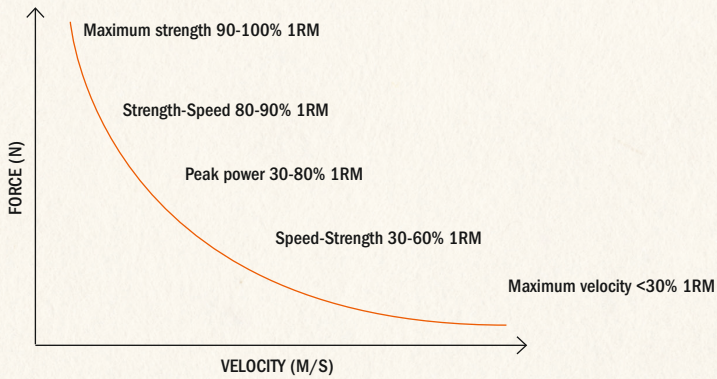
Speed–Strength Training

Training in this zone means we train speed, but with weight! The goal is to move faster than we do during Strength–Speed Training, so examples include plyometric drills like high hurdle jumps and light-loaded jump squats using 30–80% of your 1RM.

Maximal Velocity Training

Now we want to move the body, muscles and joints as fast as possible. This type of training can be very sport specific, but let's take sprinting as an example. Running 100 metres as fast as you can will represent 100% of your maximum speed, but perform that same 100-metre sprint downhill and therefore assisted (known as 'supramaximal sprinting') and you're now performing above your 100% maximum speed. This training zone is classified as being at less than 30% of your 1RM.

The Force-Velocity Curve



SPEED, FORCE AND WEIGHT EXPLAINED

So, what part of the graph should you train in?

The answer is all of them. These different training zones are just guidelines to various intensities and can be manipulated to fit the athlete in hand. They have been developed by coaches for educational purposes in order to show the effects different exercises and intensities have on athletic performance.

How long athletes spend at each training stage depends on three things:

- **Training age:** (As we know) if you're a newbie to training you have to spend more time improving strength before you refine your speed. This is because strength is highly correlated to power and improvements in strength for a beginner can also improve speed.
- **Specific sport:** Your choice of sport will determine what strength/speed qualities you need. This is just common sense, but a rugby player will need more strength work than a football player. But that doesn't mean a footballer shouldn't improve their strength and part of the strength and conditioning coach's job is to analyse and programme an athlete's training accordingly.
- **Time of season:** This will again change from sport to sport and athlete to athlete, but commonly during the preparatory period, a stronger emphasis should be on developing strength. Then as the season continues, a more specific approach should be taken and a greater emphasis placed on speed.

But (again) all athletes should train along the force–velocity curve.

Training notes

- The sessions of this workout are biomechanically divided. This means your 'push', 'pull' and 'lower-body' movements are performed in different sessions.
- Each workout uses the primary powerlifting movements to form the 'backbone' of the programme.
- The number of repetitions is kept low with plenty of rest between sets.
- This allows enough time to recover, so each movement is performed fast and is as powerful as possible.

Regarding the routine itself:

- The 3-Day Powerlifting/Ballistic Training Hybrid Programme:
  - Is performed weeks 1–4.
  - Combines conventional strength training with ballistic throws and jumps.
- The 3-Day Powerlifting/Resistance Band Hybrid Programme:
  - Is performed weeks 5–8.
  - Combines conventional strength training with resistance bands.
- The 3-Day Powerlifting /Olympic Lifting Hybrid Programme:
  - Is performed weeks 5–8.
  - Combines conventional strength training with Olympic Lifts.

This workout is ideal for any athlete wanting a solid strength and speed conditioning routine.

	3-DAY ROUTINE	3-DAY ROUTINE	3-DAY ROUTINE
	WEEKS 1–4	**WEEKS 5–8**	**WEEKS 9–12**
KEY FOCUS	Practise technique Combine conventional strength training with ballistic throws and jumps	Practise technique Combine conventional strength training with resistance bands	Practise technique Combine conventional strength training with Olympic Lifts
WORKOUT	**Monday:** Push **Tuesday:** REST **Wednesday:** Lower Body **Thursday:** REST **Friday:** Pull **Saturday:** REST **Sunday:** REST	**Monday:** Push **Tuesday:** REST **Wednesday:** Lower Body **Thursday:** REST **Friday:** Pull **Saturday:** REST **Sunday:** REST	**Monday:** Push **Tuesday:** REST **Wednesday:** Lower Body **Thursday:** REST **Friday:** Pull **Saturday:** REST **Sunday:** REST
MAIN GOAL	Refine and improve technique and increase rate of force development (ROFD) by using a powerlifting/ ballistic hybrid programme.	Refine and improve technique and increase ROFD by using a powerlifting/resistance band hybrid programme that embraces progressive resistance.	Refine and improve technique and increase ROFD by using a powerlifting/Olympic Lifting hybrid programme.

POWERLIFTING/BALLISTIC HYBRID PROGRAMME | 3-DAY ROUTINE

This workout is a fusion of strength and speed genius.

It takes inspiration from the above research and blends it with my experiences training at Brunel and Loughborough as well as the time I'd spent under the iron in Leeds with Andy Bolton. It combines traditional strength training with speed training and includes *Maximum Strength Training*, *Strength–Speed Training*, *Peak Power Training*, *Speed–Strength Training and Strength-Speed Training*.

All to bring about neuro-musculoskeletal adaptations that make you a quicker and more powerful athlete.

WEEKS 1–4 | MONDAY: PUSH

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Medicine Ball Throw	5	3	120 secs	30–60%	Speed-Strength
Plyometric Push-up	8	5	120 secs	30–80%	Peak Power
Bench Press	5	5	120 secs	90%	Strength
Push Press	8	3	120 secs	80–90%	Strength-Speed

WEEKS 1–4 | WEDNESDAY: LOWER BODY

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Depth Jump	5	3	120 secs	30–60%	Speed-Strength
Weighted Jump Squat	8	5	120 secs	30–80%	Peak Power
Squat	5	5	120 secs	90%	Strength
(Sled) Prowler Push	8	3	120 secs	80–90%	Strength-Speed

WEEKS 1–4 | FRIDAY: PULL

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Tyre Flip	5	3	120 secs	30–60%	Speed-Strength
Backward Medicine Ball	8	5	120 secs	30–80%	Peak Power
Deadlift	5	5	120 secs	90%	Strength
Dynamic Bent-over Row	8	3	120 secs	80–90%	Strength-Speed

POWERLIFTING/RESISTANCE BAND HYBRID PROGRAMME | 3-DAY ROUTINE

We are now ready to wield the power of resistance bands.

Born from knowledge and some elasticated rubber, this workout has been created to offer a different form of progressive resistance to your training to help overcome plateaus in power, strength and speed by ultimately increasing your ROFD by loading the muscles above their habitual level in new ways.

The bands can also be used to train far more functionally than conventional weight training alone.

WEEKS 5–8 | MONDAY: PUSH

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Resistance Band Push-up	8	3	120 secs	80–90%	Strength-Speed
Bench Press	5	5	120 secs	90%	Strength
Banded Fly	12	3	120 secs	80%	Strength*
Banded Front Delt Raise	12	3	120 secs	80%	Strength*
Resistance Band Tricep Pushdown	8	3	120 secs	80–90%	Strength-Speed*

WEEKS 5–8 | WEDNESDAY: LOWER BODY

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Weighted Vest Box Jump	8	5	120 secs	30–80%	Peak Power
Leg Press	5	5	120 secs	90%	Strength
Weighted Lunge	10	5	120 secs	80%	Strength
Band Resisted Squat	8	3	120 secs	80–90%	Strength*

WEEKS 5–8 | FRIDAY: PULL

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Band Assisted Pull-up	5	3	120 secs	30–60%	Strength*
Deadlift	5	5	120 secs	90%	Strength
Band Resisted Barbell Curl	12	3	120 secs	80%	Strength*
Band Resisted Single Bicep Curl	12	3	120 secs	80%	Strength*

* Progressive resistance

POWERLIFTING/OLYMPIC LIFTING HYBRID PROGRAMME | 3-DAY ROUTINE

Olympic lifts have been included in this workout for two reasons:

1. To highlight any weak links in the kinetic chain

For example, you might be able to grind out a deadlift, maybe even a squat, but the reality is you can't do this with a snatch and all the joints and muscles must be firing at their optimal level to complete one.

2. Prepare the body with neural recruitment/activation

Basically, the high level of muscle motor recruitment needed for an Olympic lift preps the body, muscle and joints to then lift big during a deadlift or squat.

You can of course design a programme that will solely focus on improving your clean and jerk and snatch, but the beauty of this particular workout is that the weight you lift is less important than the training response we're trying to achieve: to make you a faster and more powerful version of yourself.

WEEKS 9–12 | MONDAY: PUSH

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Split Jerk	5	3	120 secs	30-60%	Speed-Strength
Bench Press	5	5	120 secs	90%	Strength
Standing Barbell Press	12	3	120 secs	80%	Strength
Resistance Band Tricep Pushdown	12	3	120 secs	80%	Strength*

WEEKS 9–12 | WEDNESDAY: LOWER BODY

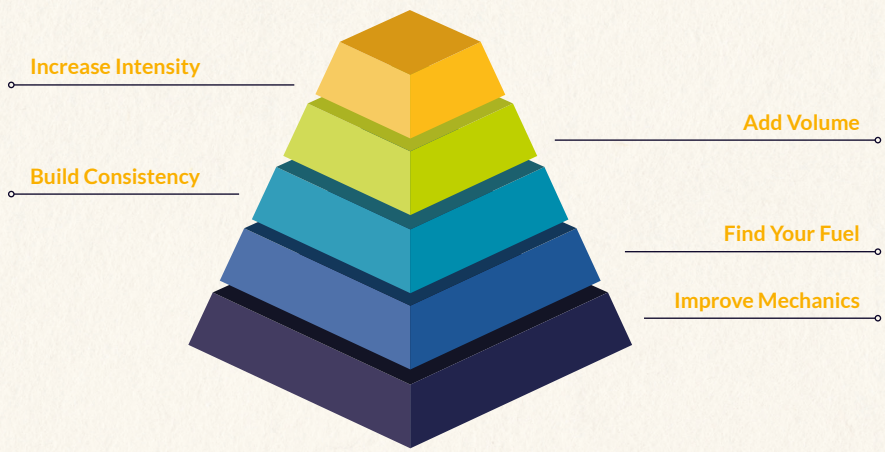
Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Depth Jump	5	3	120 secs	30–60%	Speed-Strength
Front Squat	5	5	120 secs	90%	Strength
Weighted Jump Squat	8	5	120 secs	30–80%	Peak Power
Overhead Lunge	12	3	120 secs	80%	Strength

WEEKS 9–12 | FRIDAY: PULL

Exercise	Reps	Sets	Rest	% One-Rep Max	Focus
Snatch	5	3	120 secs	30–60%	Speed-Strength
Deadlift	5	5	120 secs	90%	Strength
Jump Shrug	5	3	120 secs	30–60%	Speed-Strength
Band Assisted Pull-up	5	3	120 secs	30–60%	Strength*
Band Resisted Barbell Curl	12	3	120 secs	80%	Strength*

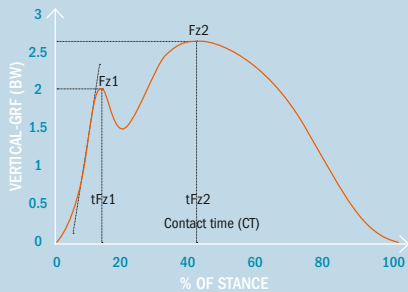
* Progressive resistance

The Endurance Pyramid of Priority

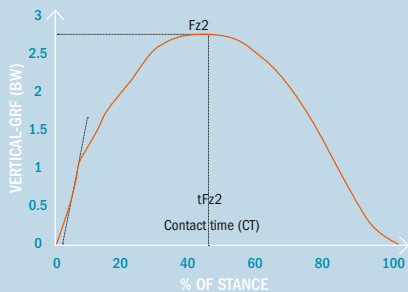


THE ART OF ENDURANCE | FOOT PLACEMENT

Heelstrike landing



Non-heelstrike landing



THE ART OF ENDURANCE | FOOT PLACEMENT

Here are some of the many errors in running technique crimes that I've committed over the years. Under no circumstances should you try any of them at home:

1. Dropped shoulders

2. Twisted body (showing twist to the left)

3. Narrow step width

4. Femur twists inwards

5. Swing leg facing outwards

6. Knees are deeply flexed

7. Leg lands too far in front

8. Low waist and lower back

9. Body leans forward

10. Restricted range of motion in hip joints

11 Lower leg tilts significantly

12. Toes point outwards after pushing off

MASTERING ‘FASTER (ELEPHANT) FEET’

YOUR 20-WEEK MARATHON GUIDE

Training notes

The keys to this programme are:

- All units are listed in kilometres.
- The hardest and most intense training period is between weeks 12 and 17.
- Numbers that are in brackets indicate the run should be done at a pace higher than your anaerobic threshold pace.
- Numbers in red indicate you should run at race pace.
- The final long-distance run should be done two weeks prior to the race.
- Two days before the race, jog 2–3 km and increase speed over 1 km to simulate the race and stimulate the legs.

WEEK		MON	TUES	WED	THUR	FRI	SAT	SUN	DISTANCE PER WEEK (KM)
1	30-minute run (walk if necessary to acclimatise the body to the distance).		10		10		10		N/A
2	Jog at a slow and comfortable pace.		3	3	3		3		12
3			3	3	3		5		14
4	Run above your anaerobic threshold pace once per week.		5		5		5		15
5			5	5	5		5		20
6	Run faster than your anaerobic threshold pace twice a week. Perform a long-distance run on Sat/Sun that is slower than your anaerobic threshold pace.		(5)		(5)		7		17
7			(5)		(5)		10		20
8			(7)		(7)		10		24
9			(7)		(7)		12		26
10			(7)		(7)		15		29
11			(7)		(5)		15		27
12	Run faster than your anaerobic threshold pace once a week. Perform a long-distance run on Sat/Sun that is slower than your anaerobic threshold pace. On the other days, jog at a slow and comfortable pace.		5	(10)			20		35
13			5	(10)			25		40
14			5	(10)			15		30
15			5	(10)		5	**25**		45
16			5	(10)			15		30
17			5	(10)		5	**30**		50
18	Run at race pace, but be sure to avoid the Exhaustion Phase.			5	10		15		30
19			7		5		5		17
20			5		3	(1)		RACE	9

THE PERFECT ROW

THE 1300-REPETITION ROWING WORKOUT

Presenting the 1300-repetition workout forged under the roof of the Cambridge Boathouse. Created to develop this concept of enduring strength, it's a workout that's become renowned at the club. Performed from start to finish with absolutely no rest, the large, full-body, compound movements performed with a high number of repetitions closely mimic the physical race demands.

BENCH PULL
Reps: 150
% One-rep max: 50%

SEATED ROW
Reps: 100
% One-rep max: 50%

SQUAT
Reps: 200
% One-rep max: 50%

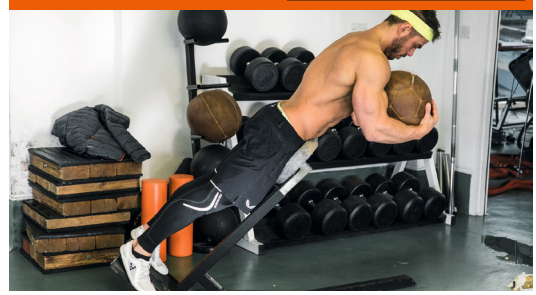
DUMBBELL SQUAT
Reps: 100
% One-rep max: 50%

STEP-UP
Reps: 200
% One-rep max: 50%

RUSSIAN TWIST
Reps: 50
% One-rep max: 50%

WEIGHTED SIT-UP Reps: 50 | % One-rep max: 50%

DORSAL RAISE Reps: 50 | % One-rep max: 50%



BENCH PRESS Reps: 100 | % One-rep max: 50%

UPRIGHT ROW Reps: 100 | % One-rep max: 50%

SINGLE BENT-OVER ROW Reps: 150 | % One-rep max: 50%

DUMBBELL PULL-OVER Reps: 50 | % One-rep max: 50%

TOTAL REPS: 1300

'Think of aerobics plus weightlifting minus the music or camaraderie. Combine unalloyed endurance with straightforward strength and demand poise, timing, and practiced form as well. Think of pure pain: that's the ergometer.'

BARRY STRAUSS, Rowing Against the Current